Statistical Mechanics



Department of Materials Science - École Polytechnique Fédérale de Lausanne

Author: Arthur Aimone

2024/2025

Contents

1 Prerequisites **3**

1.1 Fundamentals of Classical Thermodynamics . . . 3

1.1.1 Introduction and Entropy . . . 3

1.1.2 Legendre Transform and Thermodynamic Potentials . . . 6

1.1.3 A Variational Principle for the Gibbs Free Energy G . . . 7

1.1.4 Specific Heats . . . 8

1.2 Probability and Measure Theory . . . 8

2 Fundamentals of Statistical Mechanics **12**

2.1 A Statistical Definition of Entropy . . . 12

2.1.1 Equiprobability Axiom and Density of States . . . 13

2.1.2 Definition of Entropy and some Properties . . . 14

2.2 Statistical Properties of some Ensembles . . . 16

2.2.1 The Partition Function . . . 16

2.2.2 Example - A Paramagnetic Material . . . 19

2.3 The Gibbs Entropy . . . 21

2.4 Some Ensembles in the Context of MaxEnt . . . 22

2.4.1 The Microcanonical Ensemble . . . 23

2.4.2 The Canonical Ensemble . . . 23

2.4.3 The Grand-Canonical Ensemble . . . 24

2.4.4 Isothermal-Isobaric Ensemble . . . 25

2.5 System Fluctuations . . . 26

2.5.1 Particle Number Fluctuations . . . 26

2.5.2 Uncorrelated Fluctuations . . . 27

2.5.3 Ideal Gas Law . . . 28

2.5.4 Example - Langmuir Adsorption Isotherm . . . 28

2.6 Independent and Indistinguishable Particles (Quantum Approach) . . . 29

2.6.1 Independent Particles . . . 30

2.6.2 Indistinguishability . . . 31

2.6.3 Canonical Partition Functions . . . 33

2.6.4 Grand-Canonical Partition Functions - Bose-Einstein, Fermi Statistics and Boltzmann Statistics . . . 33

3 Models in Statistical Mechanics and Applications in Materials Science **36**

3.1 Vibrations and Electronic Excitations in Solids . . . 36

3.1.1 The Debye Solid . . . 38

3.1.2 Thermal Properties of Graphene . . . 38
3.1.3 The Free Electron Gas . . . 39
3.2 Simulation and Sampling (Not Examinable) . . . 40
3.3 Ising Model . . . 43
3.3.1 Ising Model in 1-Dimension . . . 43
3.3.2 Ising Model in 2-Dimensions and Above - The Mean Field Model . . . 44
3.3.3 Magnetization . . . 45
3.3.4 Susceptibility . . . 45
3.3.5 Final Remarks and Mappings . . . 46
3.4 Statistical Mechanics of Liquids . . . 46
3.4.1 Classical Methods, Reduced Distribution Functions and the Pair Correlation Function . . . 46
3.4.2 An Application - Scattering from a Liquid . . . 48
3.4.3 Reversible Work Theorem . . . 48
3.4.4 Pair Potentials and Thermodynamic Properties of Liquids . . . 49
3.5 Statistical Mechanics of Polymers . . . 50
3.5.1 Characterizing Polymer Configurations and some General Relations . . . 51
3.5.2 Freely Jointed Chain . . . 52
3.5.3 Freely Rotating Chain . . . 52
3.5.4 Partition Function at Constant f and Polymer Extension . . . 53

These lecture notes are based on the *Statistical Mechanics* course given at EPFL for the academic year 2024/2025 by Prof. Michele Ceriotti.

Chapter 1

Prerequisites

1.1 Fundamentals of Classical Thermodynamics

In this chapter, whenever we write partial derivatives in the form $\left(\frac{\partial f}{\partial u}\right)_{\mathbf{X}}$ we mean a partial derivative of f with respect to u evaluated at constant $\mathbf{X}$.

1.1.1 Introduction and Entropy

The task of measuring position and velocities of a microscopic amount of atoms or molecules is practically impossible. Instead, we may characterize the macroscopic state of *equilibrium* systems by a small number of macroscopic observables, e.g. volume V, temperature T, pressure ρ or molar amounts of components n_i.

Definition 1.1.1. ***State variables*** *are macroscopic quantities that only depend on the state of the system and not on how the system has been prepared (e.g. T, ρ, E, V, n,...).*

Infinitesimal changes in state variables are exact differentials (e.g. $\int_{V_1}^{V_2} dV = V_2 - V_1$, irrespective of the path). Extensive variables (n, V) double when system size is doubled, whereas intensive variables do not depend on system size (e.g. T, ρ).
Concerning boundary conditions, the boundary of the system can be open or closed relative to the exchange of mass, and conducting or adiabatic relative to exchange of heat with its surroundings (closed + adiabatic = isolated). In this context, adiabatic relates to a process in which heat does not enter or leave the system concerned.

Definition 1.1.2. *The* ***internal energy*** *is a function of state that expresses the potential the system has to generate work. It can be defined by assuming it is:*
(1) Extensive ($E = E_1 + E_2$),
(2) Conserved. It can change by doing work or by exchanging heat.

$$dE = đW + đQ \qquad \textbf{\textit{First Law}} \tag{1.1}$$

Here, d denotes an exact differential and $đ$ denotes an inexact differential as it depends on the path followed during a transformation. E is the energy, W the work and Q is the heat.

Work is associated with change of one of the extensive mechanical variables X defining the state of the system. For each variable there is a (generalized) force f that links it with the work done by changing it in general by $đW = fdX$ (e.g. $đW = -\rho_{\text{ext}}dV$). We can measure dE as the work spent doing an adiabatic transformation between two states A and B.
We may consider the first law (1.1) as the definition of heat:
$\rightarrow$ Do an adiabatic transformation to measure the change in internal energy $E_B - E_A$.
$\rightarrow$ Repeat with open walls, measure the new work $đW$ and obtain $đQ$ as the difference:

$$đQ = dW_{\text{ad}} - đW. \tag{1.2}$$

Definition 1.1.3. ***Equilibrium*** *states do not change unless perturbed. If the system returns to a state after a small perturbation, it is said to be stable.*

Definition 1.1.4. *An* ***equation of state*** $f(\mathbf{X}) := f(E, V, n_i, ...) = 0$*, determines which values of thermodynamic variables are compatible with equilibrium.*

Reversible transformations proceed along a series of equilibrium states. On the other hand, irreversible transformations traverse values of thermodynamic variables that violate the equation of state.
Interesting questions arise when we are searching for equilibrium states: How do we determine if a system is in equilibrium? How do we determine what is the equilibrium state of a system? How do we determine the evolution of a system after we remove some internal constraints? We can try to define a single state function to answer all these questions: entropy!

Definition 1.1.5 (Axiomatic Definition of Entropy)**.** *We define the* ***entropy*** $S(E, \mathbf{X})$ *as:*
(1) An extensive function of state.
(2) A monotonically increasing function of E, where we define the temperature as:

$$\left(\frac{\partial E}{\partial S}\right)_{\mathbf{X}} = \left[\left(\frac{\partial S}{\partial E}\right)_{\mathbf{X}}\right]^{-1} = T > 0.$$

If it is possible to obtain state B adiabatically from state A, then $S(B) \geq S(A)$. This condition $\Delta S_{\text{ad}} \geq 0$ is the **Second Law**. Furthermore, if a process is reversible, then one can also reach adiabatically A from B and it implies that $S(B) = S(A)$ (since $S(B) \geq S(A)$ and also $S(B) \leq S(A)$).
We now look at the relation between between entropy and reversible heat. We consider a system defined by S, E and mechanical variables $\mathbf{X}$. For a reversible process (rev. process) one can cast the equation of state into $S = S(E, \mathbf{X})$. We can now compare the differentials of S and E. For S we find:

$$dS = \left(\frac{\partial S}{\partial E}\right)_{\mathbf{X}} dE + \sum_i \left(\frac{\partial S}{\partial X_i}\right)_{E,\mathbf{X}'_i} dX_i = \frac{1}{T} dE + \sum_i \left(\frac{\partial S}{\partial X_i}\right)_{E,\mathbf{X}'_i} dX_i. \tag{1.3}$$

For E, using the First Law, we find:

$$dE = đQ_{\text{rev.}} + đW_{\text{rev.}} = đQ_{\text{rev.}} + \underbrace{\sum_i \frac{\partial E}{\partial X_i} dX_i}_{\text{rev. work}}. \tag{1.4}$$

- In particular, for an adiabatic, reversible process we have $dS = 0$ and $đQ = 0$ hence:

$$0 = dS = \frac{1}{T}dE + \sum_i \left(\frac{\partial S}{\partial X_i}\right)_{E,\mathbf{X}'_i} dX_i = \frac{1}{T}\left(đQ_{\text{rev.}} + \sum_i \frac{\partial E}{\partial X_i}dX_i\right) + \sum_i \left(\frac{\partial S}{\partial X_i}\right)_{E,\mathbf{X}'_i} dX_i$$
$$= \sum_i \frac{1}{T}\frac{\partial E}{\partial X_i}dX_i + \sum_i \left(\frac{\partial S}{\partial X_i}\right)_{E,\mathbf{X}'_i} dX_i.$$

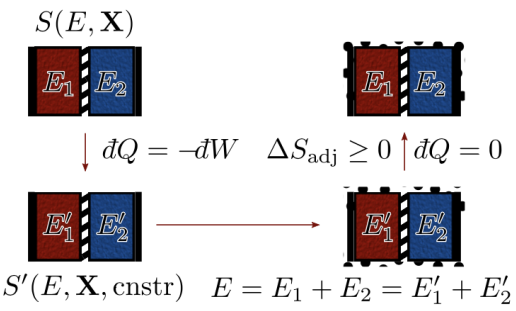
Now, since every term in the sums are positive we finally obtain:

$$\sum_i \left(\frac{1}{T}\frac{\partial E}{\partial X_i} + \left(\frac{\partial S}{\partial X_i}\right)_{E,\mathbf{X}'_i}\right) dX_i = 0 \Longrightarrow \left(\frac{\partial E}{\partial X_i}\right)_{S,\mathbf{X}'_i} = -T\left(\frac{\partial S}{\partial X_i}\right)_{E,\mathbf{X}'_i}.$$

- Now, for a reversible but non-adiabatic process, we obtain Clausius Principle stating that $TdS = đQ_{\text{rev.}}$ and therefore we obtain the following differential:

$$dE = TdS + \sum_i \frac{\partial E}{\partial X_i}dX_i.$$

Consider now a system to which we applied an internal constraint by, for example, fixing the partitioning of E between two subsystems. We start in equilibrium and make an arbitrary transformation that only changes the partitioning of energy between the two subsystems. This gives a state with a different entropy S' but same energy $E = E_1 + E_2$, where from the First Law we have $đQ + đW = 0$. We then relax back to equilibrium adiabatically, where from the axioms we have $S' < S$.
The equilibrium partitioning is the one that maximizes entropy: $\Delta S_{\text{adj.}} \geq 0$ results in a general maximum-entropy principle (MaxEnt principle).



Under the conservation conditions $E = E_1 + E_2$, $\mathbf{X} = \mathbf{X}_1 + \mathbf{X}_2$ and given the MaxEnt principle:

$$S(E, \mathbf{X}, \text{constraint}) = S(E_1, \mathbf{X}_1) + S(E_2, \mathbf{X}_2) \leq S(E, \mathbf{X}). \tag{1.5}$$

Recall that $\frac{\partial S}{\partial E} > 0$, then if $S(E', \mathbf{X}) = S(E, \mathbf{X}, \text{constraint})$ it must be that $E' < E$. That is, introducing constraints at constant entropy necessarily implies an increase of internal energy.
We now look at the thermal equilibrium and convergence. If we introduce infinitesimal constraints δY around equilibrium, we can express this condition as:

$$\delta E_{S,\mathbf{X}} = E(S, \mathbf{X}, \delta Y) - E(S, \mathbf{X}) \approx \left(\frac{\partial E}{\partial Y}\right)_{S,\mathbf{X},\delta Y = 0}, \quad \delta Y \geq 0. \tag{1.6}$$

Now, what happens if we tried to partition energy between the two subsystems, at constant total energy?

$$\underbrace{0 \geq \delta S_{E,\mathbf{X}}}_{\text{MaxEnt Principle}} = \underbrace{\delta S_1 + \delta S_2}_{\text{Extensive Entropy}} = \left(\frac{\partial S_1}{\partial E_1}\right)_{\mathbf{X},\delta E_1=0} \delta E_1 + \left(\frac{\partial S_2}{\partial E_2}\right)_{\mathbf{X},\delta E_2=0} \delta E_2.$$

Due to the constant energy condition, $\delta E_2 = -\delta E_1$, and considering that $\frac{\partial S}{\partial E} = \frac{1}{T}$ we find:

$$\frac{1}{T_1}\delta E_1 + \frac{1}{T_2}\delta E_2 = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\delta E_1 \leq 0, \quad \text{for all } \delta E_1. \tag{1.7}$$

This implies that $T_1 = T_2$.
Now consider an initial state $T_1 \neq T_2$. An adiabatic spontaneous evolution will have $\Delta S_1 + \Delta S_2 = \Delta S > 0$, and:

$$\left(\frac{\partial S_1}{\partial E_1}\right)_{\mathbf{X}} \Delta E_1 + \left(\frac{\partial S_2}{\partial E_2}\right)_{\mathbf{X}} \Delta E_2 > 0 \Longrightarrow \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\Delta E_1 > 0. \tag{1.8}$$

The MaxEnt Condition implies that energy flows from the hot to the cold subsystem:

$$T_2 > T_1 \Longrightarrow \frac{1}{T_2} < \frac{1}{T_1} \Longrightarrow \Delta E_1 > 0. \tag{1.9}$$

1.1.2 Legendre Transform and Thermodynamic Potentials

We first start with a practical mathematical tool. Consider a function of N variables $f(x_1, ..., x_N)$. One can write its differential using the chain rule as:

$$df = \sum_i \frac{\partial f}{\partial x_i} dx_i. \tag{1.10}$$

Say we wanted to define a function, related to f, that instead has a differential form that depends on another set of variables u_i. Consider $g := f - \sum_i x_i u_i$ then:

$$dg = \sum_i \frac{\partial f}{\partial x_i} dx_i - \sum_i u_i dx_i - \sum_i x_i du_i. \tag{1.11}$$

And now, if we set $u_i := \frac{\partial f}{\partial x_i}$, we may cancel the differentials dx_i. To this end, we obtain the following tool:

Definition 1.1.6. *The **Legendre transform** of f is given by:*

$$g(u_1, ..., u_N) := f - \sum_i x_i \frac{\partial f}{\partial x_i}. \tag{1.12}$$

Note that the transform can be applied to a subset of the x_i's.
The most important term of dE in terms of mechanical boundary conditions is $-\rho dV$, that can be supplemented by composition variables. E is a natural function of S, V, n_i and so introducing the chemical potential $\mu_i = (\partial E/\partial n_i)_{S,V,n_i}$ the differential becomes:

$$dE = TdS - \rho dV + \sum_i \mu_i dn_i. \tag{1.13}$$

Now, by Legendre transform specific variables we obtain different well-defined quantities.

Definition 1.1.7. *If we Legendre transform the entropy S we obtain the* ***Helmoltz free energy*** *A, given by:*

$$A = E - TS. \tag{1.14}$$

In particular, its differential is given by:

$$dA = dE - TdS - SdT = TdS - \rho dV + \sum_i \mu_i dn_i - TdS - SdT = -\rho dV + \sum_i \mu_i dn_i - SdT.$$

Definition 1.1.8. *If we Legendre transform the volume V we obtain the* ***enthalpy*** *H, given by:*

$$H = E + \rho V. \tag{1.15}$$

In particular, its differential is given by:

$$dH = dE + \rho dV + Vd\rho = TdS - \rho dV + \sum_i \mu_i dn_i + \rho dV + Vd\rho = TdS + \sum_i \mu_i dn_i + Vd\rho.$$

Definition 1.1.9. *If we Legendre transform the volume V and the entropy S we obtain the* ***Gibbs free energy*** *G, given by:*

$$G = E - TS + \rho V. \tag{1.16}$$

In particular, its differential is given by:

$$\begin{aligned} dG &= dE - TdS - SdT + \rho dV + Vd\rho \\ &= TdS - \rho dV + \sum_i \mu_i dn_i - TdS - SdT + \rho dV + Vd\rho = \sum_i \mu_i dn_i - SdT + Vd\rho. \end{aligned}$$

1.1.3 A Variational Principle for the Gibbs Free Energy G

Consider a composite system $A+B$ enclosed in a temperature/pressure reservoir R, at T^R and ρ^R. Using adiabaticity and a constant volume condition we have: $d(E^A + E^B + E^R) = 0$ and $d(V^A + V^B + V^R) = 0$. Furthermore, with an equilibrium condition $d(S^A + S^B + S^R) = 0$ we have:

$$dS^A + dS^B = -dS^R, \;\; dV^A + dV^B = -dV^R, \;\; dE^A + dE^B = -dE^R.$$

Now using the differential $dE = TdS - \rho dV$ and the constant energy E:

$$\begin{aligned} 0 &= T^A dS^A + T^B dS^B - \rho^A dV^A - \rho^B dV^B - \rho^R dV^R + T^R dS^R \\ &= (T^A - T^R)dS^A + (T^B - T^R)dS^B - (\rho^A - \rho^R)dV^A - (\rho^2 - \rho^R)dV^B. \end{aligned}$$

An equilibrium condition implies that $T^A = T^B = T^R = T$ and $\rho^A = \rho^B = \rho^R = \rho$. Hence, considering the $A+B$ subsystem and remembering that $dG = d(E - TS + \rho V)$, with the T and ρ constraint:

$$\begin{aligned} 0 = dE^A + dE^B + dE^R &= dE^A + dE^B + TdS^R - \rho dV^R \\ &= d(E^A + E^B) - Td(S^A + S^B) + \rho d(V^A + V^B) = \Big(d(G^A + G^B)\Big)_{T,\rho}. \end{aligned}$$

1.1.4 Specific Heats

Considering double derivatives of thermodynamic potentials reveals some Maxwell-type relations between macroscopic quantities:

$$\frac{\partial^2 E}{\partial V \partial S} = \left(\frac{\partial T}{\partial V}\right)_S \text{ and, } \frac{\partial^2 E}{\partial S \partial V} = -\left(\frac{\partial \rho}{\partial S}\right)_V \Longrightarrow \left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial \rho}{\partial S}\right)_V.$$

Similar equations hold for other potentials. With $\frac{\partial^2 H}{\partial S \partial \rho}$ we obtain:

$$\left(\frac{\partial T}{\partial \rho}\right)_S = \left(\frac{\partial V}{\partial S}\right)_\rho. \tag{1.17}$$

With derivatives of the Helmoltz free energy $\frac{\partial^2 A}{\partial T \partial V}$ we obtain:

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial \rho}{\partial T}\right)_V. \tag{1.18}$$

And finally, with derivatives of the Gibbs free energy $\frac{\partial^2 G}{\partial T \partial \rho}$ we have:

$$-\left(\frac{\partial S}{\partial \rho}\right)_T = \left(\frac{\partial V}{\partial T}\right)_\rho. \tag{1.19}$$

1.2 Probability and Measure Theory

In this section, we present a list of definitions and properties in Probability and Measure Theory.

Definition 1.2.1. *A **probability** is a function $\mathbb{P} : \mathcal{E} \longrightarrow \mathbb{R}^+$ that satisfies the following axioms:*
(1) For all $A \in \mathcal{E}$, $\mathbb{P}(A) \geq 0$.
(2) $\mathbb{P}(\Omega) = 1$.
(3) For any countable collection of sets $\{A_i\}$ that are non-overlapping (i.e. such that $A_i \cap A_j = \varnothing$):

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i). \tag{1.20}$$

Proposition 1.2.2. *If $\overline{A}$ is the complement of A (i.e. $A \cup \overline{A} = \Omega$ and $A \cap \overline{A} = \varnothing$):*

$$\mathbb{P}\left(\overline{A}\right) = 1 - \mathbb{P}(A). \tag{1.21}$$

Proof. Considering axiom 3, and using the fact that A and $\overline{A}$ are non-overlapping we have that $\mathbb{P}\left(A \cup \overline{A}\right) = \mathbb{P}(A) + \mathbb{P}\left(\overline{A}\right)$. Since $A \cup \overline{A} = \Omega$ and with axiom 2, $\mathbb{P}(\Omega) = 1$ we finally have:

$$1 = \mathbb{P}(\Omega) = \mathbb{P}\left(A \cup \overline{A}\right) = \mathbb{P}(A) + \mathbb{P}\left(\overline{A}\right) \Longrightarrow \mathbb{P}\left(\overline{A}\right) = 1 - \mathbb{P}(A). \tag{1.22}$$

□

Proposition 1.2.3. *The empty set has probability 0,* $\mathbb{P}(\varnothing) = 0$.

Proof. Remark that $A = A \cup \varnothing$ and also that A and $\varnothing$ are non-overlapping. Using Axiom 3, we obtain:

$$\mathbb{P}(A) = \mathbb{P}(A \cup \varnothing) = \mathbb{P}(A) + \mathbb{P}(\varnothing) \Longrightarrow \mathbb{P}(\varnothing) = 0. \tag{1.23}$$

□

Definition 1.2.4. *Consider two event sets* $A \in \mathcal{E}$ *and* $B \in \mathcal{E}'$. *We can define the* ***joint probability*** $\mathbb{P}(A, B)$ *as the probability that two events compatible with the sets* A *and* B *both occur.*

$$\mathbb{P}(A, B) = \mathbb{P}(\omega \in A \ \ and \ \ \omega' \in B). \tag{1.24}$$

Definition 1.2.5. *Given a joint probability we can recover the probability of individual events by considering the* ***marginal probability****:*

$$\mathbb{P}(A, \Omega') = \mathbb{P}(A) \quad \ and \quad \ \mathbb{P}(\Omega, B) = \mathbb{P}(B). \tag{1.25}$$

Definition 1.2.6. *We define the* ***conditional probability*** *as:*

$$\mathbb{P}(A|B) = \mathbb{P}(\omega \in A \ \ knowing \ that \ \ \omega' \in B). \tag{1.26}$$

Say we could decompose Ω into a countable set of disjoint components $A_1, ...$ such that $\cup_i A_i$ and $A_i \cap A_j = \varnothing$. We can use axiom 3 to obtain marginals by summing over all possible events:

$$\mathbb{P}(B) = \mathbb{P}(\Omega, B) = \sum_i \mathbb{P}(A_i, B), \quad \mathbb{P}(\omega_2) = \sum_i \mathbb{P}(\omega_i, \omega_2), \quad \mathbb{P}(y) = \int \mathbb{P}(x, y) dx.$$

Generalizing, we can reduce a complex joint probability by a marginalization:

$$\mathbb{P}(B, C, D) = \sum_i \mathbb{P}(A_i, B, C, D), \quad \mathbb{P}(\omega_2, \omega_3) = \sum_i \mathbb{P}(\omega_i, \omega_2, \omega_3), \quad \mathbb{P}(y, z) = \int \mathbb{P}(x, y, z) dx.$$

Proposition 1.2.7 (Bayes' Formula)**.** *The conditional probability can be related to the joint probability:*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A, B)}{\mathbb{P}(B)}. \tag{1.27}$$

Definition 1.2.8. *Two events* A *and* B *are* ***independent*** *if and only if:*

$$\mathbb{P}(A, B) = \mathbb{P}(A)\mathbb{P}(B). \tag{1.28}$$

An event can often be associated with the value of one or more quantities, $X_i(\omega)$. For continuous variables, we have to define probability densities that only make sense when written in an integral form, e.g. $\mathbb{P}(X)dX$. We can also of course compute functions of random variables, e.g. $f(X)$. In fact,

functions of random variables are in turn random variables, characterized by their own distribution. As a practical tool, if $f(X)$ is monotonic, $\mathbb{P}(f)df = \mathbb{P}\left(X^{-1}(f)\right) f'(X)dX$.

Definition 1.2.9. *The* **n-*th moment*** *of a random variable is defined as:*

$$\langle X^n \rangle = \sum_X X^n \mathbb{P}(X) \quad or, \quad \langle X^n \rangle = \int X^n \mathbb{P}(X) dX. \tag{1.29}$$

Moment of high order contain information on lower moments, this is due to the fact that $\langle X^{2n} \rangle \geq \langle X^n \rangle^2$. We can also define combinations of moments that (to an extent) eliminate this dependence, called cumulants. These can be complex, but the first is equal to the average, and the second to the variance:

$$\mathrm{Var}(X) = \left\langle (X - \langle X \rangle)^2 \right\rangle = \langle X^2 \rangle - \langle X \rangle^2. \tag{1.30}$$

Definition 1.2.10. *The* ***covariance*** *of two random variables is defined as:*

$$\langle X_i, X_j \rangle = \langle (X_i - \langle X_i \rangle)(X_j - \langle X_j \rangle) \rangle = \langle X_i X_j \rangle - \langle X_i \rangle \langle X_j \rangle. \tag{1.31}$$

Remark that if the random variables are independent in pairs, then $\langle X_i, X_j \rangle = \mathrm{Var}(X_i)\delta_{ij}$.
Now, suppose we have N evaluations (may not be independent) of a random variable $X(i)$. The mean of the set of events is itself a random variable $\overline{X}_N = \frac{1}{N}\sum_i X(i)$. Its average is equal to:

$$\langle \overline{X}_N \rangle = \frac{1}{N} \sum_i \langle X(i) \rangle = \langle X \rangle. \tag{1.32}$$

Its variance depends on the correlation between different occurrences:

$$\mathrm{Var}\left(\overline{X}_N\right) = \langle \overline{X}_N, \overline{X}_N \rangle = \frac{1}{N^2} \sum_{ij} \langle X(i), X(j) \rangle. \tag{1.33}$$

Furthermore, if we assume uncorrelated samples $\langle X(i), X(j) \rangle = \mathrm{Var}(X)\delta_{ij}$ then:

$$\mathrm{Var}\left(\overline{X}_N\right) = \frac{Var(X)}{N^2} \sum_{ij} \delta_{ij} = \frac{Var(X)}{N}. \tag{1.34}$$

Remark 1.2.11. *Often, samples come from a time series and different samples are correlated so* $c_{XX}(j) = \frac{\langle X(i), X(i+j) \rangle}{\mathrm{Var}(X)}$ *will be a decaying autocorrelation function of* j*. Physically,* $c_{XX}(j)$ *says how fast fluctuations from the mean are forgotten.*

Definition 1.2.12. *Given a probability function* $\mathbb{P}(X)$ *we can define the* ***characteristic function****, which is just its Fourier transform, as:*

$$\phi(s) = \langle \exp(isX) \rangle = \int \mathbb{P}(X) e^{isX} dX. \tag{1.35}$$

If $\mathbb{P}$ is normalized, then $\phi(0) = 1$. Furthermore, from the properties of the Fourier transform, $\phi^{(n)}(0) = i^n \langle X^n \rangle$. This is why $\phi(s)$ is also known as the moment generating function.
Remark that considering the characteristic function associated with the joint sum of two independent

variables:

$$\phi_{1,2}(s) = \int \mathbb{P}(X_1, X_2) e^{is(X_1+X_2)} dX_1 dX_2 = \phi_1(s)\phi_2(s). \tag{1.36}$$

This is a very useful fact to show independence of two random variables, and also to predict the distribution of the combination of independent variables (such as the mean): For example,

$$\phi_{\overline{X}_N}(s) = \prod_i \phi_{X(i)}\left(\frac{s}{N}\right) = \left[\phi_X\left(\frac{s}{N}\right)\right]^N. \tag{1.37}$$

Theorem 1.2.13 (Central Limit Theorem - CLT)**.** *Suppose $\{X_i\}$ is a sequence of i.i.d. random variables such that $\mathbb{E}(X_i) = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then:*

$$\sqrt{n}\left(\overline{X}_n - \mu\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2\right). \tag{1.38}$$

Example 1.2.14. *Consider a distribution with finite moments, assuming for simplicity that $\mu = 0$ and $\sigma^2 = 1$. Taylor expand the characteristic function:*

$$\phi_X(s) \approx 1 - \frac{1}{2}s^2 + \mathcal{O}(s^3). \tag{1.39}$$

Now consider the characteristic function of the mean:

$$\phi_{\overline{X}_N}(s) = \left[\phi_X\left(\frac{s}{N}\right)\right]^N \approx \left[1 - \frac{1}{N}\frac{1}{2}\frac{s^2}{N} + \mathcal{O}\left(\left(\frac{s}{N}\right)^3\right)\right]^N \xrightarrow{N\to\infty} e^{-\frac{s^2}{2N}}. \tag{1.40}$$

Inverting the definition of the characteristic function, one gets $\mathbb{P}\left(\overline{X}_N\right) \propto e^{-N\overline{X}_N^2/2}$. Hence the mean is a Gaussian with a variance $\frac{1}{N}$. Note that we have used the definition of the exponential function $\lim_{N\to\infty}\left(1 + \frac{x}{N}\right)^N = e^x$.

Definition 1.2.15. *We say that a random variable X follows the **Binomial distribution**, $X \sim Bin(n,p)$, with n independent Bernoulli trials each with probability p if:*

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}. \tag{1.41}$$

For a Binomial distribution, $X \sim \mathrm{Bin}(n,p)$, we have that $\langle X \rangle = np$ and $\mathrm{Var}(X) = np(1-p)$.

Chapter 2

Fundamentals of Statistical Mechanics

In this chapter, we introduce the fundamental notions in Statistical Mechanics, comprising of the statistical definition of entropy and the description of some ensembles (e.g. canonical ensemble). We will then end on the definition of the Gibbs entropy and the study of the so-called partition function.

2.1 A Statistical Definition of Entropy

It is practically impossible to characterize fully the state of a macroscopic system. At equilibrium, systems can be comprehended by a small number of macroscopic observables, e.g. volume V, temperature T, pressure ρ or molar amounts of components n_i.

Definition 2.1.1. *A **microstate** is a complete characterization of the state of all microscopic components of a system.*

Classically, this involves assigning some positions $\mathbf{q}_i$ and momenta $\mathbf{p}_i$ for all the particles at a given time. In the view of Quantum Mechanics, it involves determining quantum numbers $|\nu\rangle$ associated with a complete set of commuting observables.
Now, imagine that we could identify the microstate of a system and measure it at different times. We could compute the mean of any observable O by summing over the values it takes for the different microstates, i.e. $\langle O\rangle = \frac{1}{N}\sum_t O_{\nu(t)}$. Classically, we may write this as $\frac{1}{N}\sum_t O(\mathbf{q}(t),\mathbf{p}(t))$ and quantum mechanically we have $\frac{1}{N}\sum_t \langle\nu(t)|\,\widehat{O}\,|\nu(t)\rangle$. The assumption here is called *ergodicity*, meaning that within the time of an observation (or given the large size of the observed sample) all representative states will be visited. Now, we can rewrite the averages looking at how often each state appears and then defining a probability distribution. Classically we now have $\int \mathbb{P}(\mathbf{q},\mathbf{p})O(\mathbf{q},\mathbf{p})d\mathbf{q}d\mathbf{p}$ and quantum mechanically we have $\sum_\nu \mathbb{P}(\nu)\,\langle\nu|\,\widehat{O}\,|\nu\rangle$.

Definition 2.1.2. *The probability defines an **ensemble**, which corresponds to a set of configurations that are consistent with the thermodynamic equilibrium conditions.*

The statistical mechanics approach is to obtain these probability distributions and to avoid the need of measuring individual microstate by predicting the average properties of a macroscopic system.

2.1.1 Equiprobability Axiom and Density of States

We need a principle to determine the probability of a state. The following proposition is the **equiprobability axiom**.

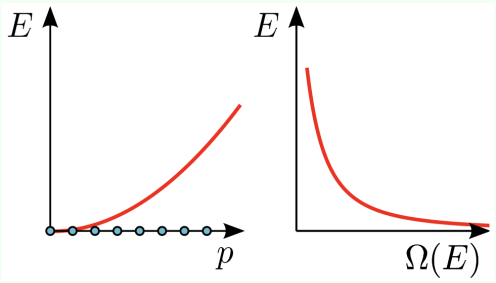
Proposition 2.1.3 (Equiprobability Axiom)**.** *For a closed and isolated system, with constant N (number of particles), V (volume) and E (energy), all the microstates that are consistent with these macroscopic observables have the same probability.*

Definition 2.1.4. *An ensemble with constant NVE is called a **microcanonical ensemble**.*

For the description of a system, we will only need to define a counter Ω for the number of states that takes into account the possible degeneracy of energy levels. Remark that for large enough quantum systems, it is convenient to consider a smooth limit of the discrete Ω and use a density of state as in the classical case. In Quantum Mechanics we will write $\Omega_E(N,V)$ (where E is the energy level) with associated probability $\mathbb{P}_{E,\nu} = \frac{1}{\Omega_E(N,V)}$, and classically we will consider $\Omega(N,V,E)dE$.

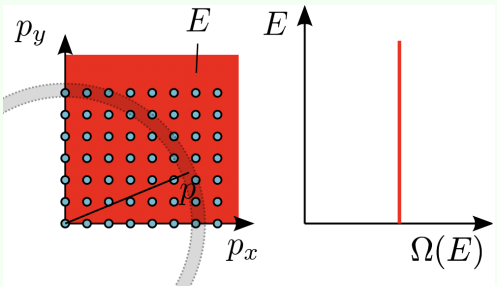
Example 2.1.5. *We now look at an example where the equiprobability axiom actually involves deep assumptions on the nature of a microstate - namely what is a complete determination of one.*
Suppose we have a system with only the kinetic energy $E = \frac{\mathbf{p}^2}{2m}$. Then, in 1-dimension, since we consider $\mathbb{P}(p) \propto 1$ and using the Kronecker-Delta function:

$$\Omega(E) \propto \int \delta\left(E - \frac{p^2}{2m}\right) dp = \int \delta(E-E')\sqrt{\frac{m}{2E'}}dE' = \sqrt{\frac{m}{2E}}. \tag{2.1}$$



In 2-dimensions (denoting $d^2\mathbf{p} = dp_x dp_y$), and considering $\mathbb{P}(p_x,p_y)$ we obtain:

$$\Omega(E) \propto \int \delta\left(E - \frac{p^2}{2m}\right) d^2\mathbf{p} = 2\pi \int \delta\left(E - \frac{p^2}{2m}\right) pdp = 2\pi \int \delta(E-E')\sqrt{\frac{m}{2E'}}\sqrt{2mE'}dE' = 2\pi m. \tag{2.2}$$



Example 2.1.6 (The 3D Particle in a Box)**.** *As a 3-dimensional example, we consider the quantum problem of a particle in a box. To this end, consider a particle in a 3-dimensional box of size L. Each state is characterized by three quantum numbers $|n_x n_y n_z\rangle$ and has an energy $E \propto \frac{\hbar^2 \mathbf{k}^2}{2m}$ with $k_{x,y,z} = \frac{n_{x,y,z}}{L}$ (n being the molar mass). As usual, considering $\mathbb{P}(\mathbf{k}) \propto 1$:*

$$\Omega(E) = \sum_{n_x,n_y,n_z} \delta\left(E - E_{n_x n_y n_z}\right) \propto \int \delta\left(E - \frac{\hbar^2 k^2}{2m}\right) d\mathbf{k}. \tag{2.3}$$

Now, for the change of variable, we use the facts that $d\mathbf{k} = 4\pi k^2 dk$, $dE = \frac{\hbar^2 k}{m} dk$ and $k = \sqrt{\frac{2Em}{\hbar^2}}$. Rearranging the second equation and multiplying both sides by k yields $k^2 dk = \frac{m}{\hbar^2} k dE$. Now, substituting this into the first equation gives $d\mathbf{k} = 4\pi(k^2 dk) = 4\pi m k dE$. Therefore, inputting in the formula for k yields $d\mathbf{k} = 4\pi m\sqrt{\frac{2Em}{\hbar^2}} dE \propto \sqrt{\frac{2Em}{\hbar^2}} dE$. Hence, only keeping terms in E gives us:

$$\Omega(E) \propto \int \delta\left(E - \frac{\hbar^2 k^2}{2m}\right) d\mathbf{k} \propto \int \delta(E - E')\sqrt{\frac{2E'm}{\hbar^2}} dE' = \sqrt{\frac{2Em}{\hbar^2}}. \tag{2.4}$$

2.1.2 Definition of Entropy and some Properties

In this section, we will use Stirling's approximation $\ln(N!) \approx N\ln(N) - N$ as $N \to \infty$ and $\frac{S}{k_B} = \ln(\Omega)$.

Definition 2.1.7. *We define **entropy** as:*

$$S(E,V,N) = k_B \ln\left(\Omega(E,V,N)\right). \tag{2.5}$$

This definition makes entropy additive. Indeed, given two non-interacting systems with number of states Ω_1 and Ω_2, the combined number of states is $\Omega = \Omega_1\Omega_2$. Then:

$$S = k_B \ln(\Omega_1\Omega_2) = k_B \ln(\Omega_1) + k_B \ln(\Omega_2). \tag{2.6}$$

Furthermore, if we consider a system B where E,V and N are known and we consider an extra condition C, then the entropy S is consistent with a MaxEnt principle:

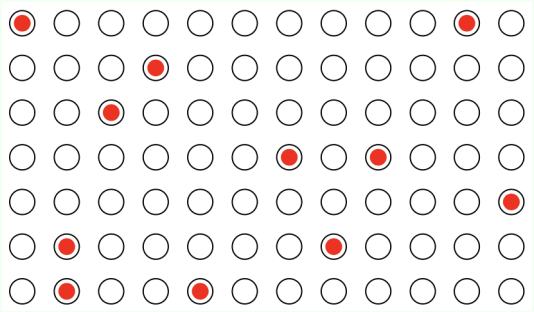
$$B(E,V,N,C) \subseteq B(E,V,N) \Longrightarrow \Omega(E,V,N,C) \leq \Omega(E,V,N) \Longrightarrow S(E,V,N,C) \leq S(E,V,N). \tag{2.7}$$

Remark 2.1.8. *Remark that the condition $\left(\frac{\partial S}{\partial E}\right)_X = \frac{1}{T} > 0$ requires:*

$$\beta = \frac{1}{k_B T} = \left(\frac{\partial \ln(\Omega)}{\partial E}\right)_X > 0. \tag{2.8}$$

Example 2.1.9 (Atoms on a Surface)**.** *Consider N absorbed atoms on a surface with M equivalent sites. All the sites are equivalent and the atoms do not interact with each other, so the energy does not depend on the distribution of the atoms:*

$$\Omega(0,M,N) = \binom{M}{N} = \frac{M!}{N!(M-N)!}. \tag{2.9}$$

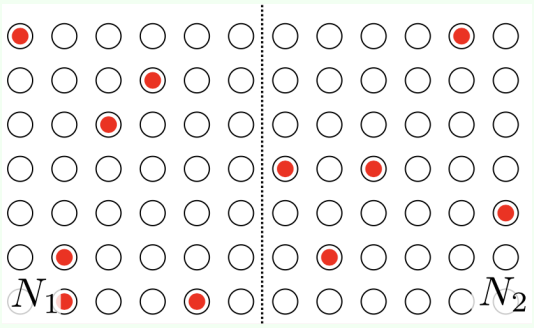


For $N, M >> 1$, we can use Stirling's approximation to get an estimate of S:

$$\frac{S}{k_B} \approx M\ln(M) - N\ln(N) - (M-N)\ln(M-N). \tag{2.10}$$

Now, partition the surface in half ($2 \times \frac{M}{2}$ sites), and fix N_1 atoms to be in region 1 and N_2 in region 2 (setting $N = N_1 + N_2$). We then have:

$$\Omega_1\left(0, \frac{M}{2}, N_1\right) = \binom{\frac{M}{2}}{N_1} = \frac{\left(\frac{M}{2}\right)!}{N_1!\left(\frac{M}{2} - N_1\right)!} \quad \textit{and,} \quad \Omega_2\left(0, \frac{M}{2}, N_2\right) = \binom{\frac{M}{2}}{N_2} = \frac{\left(\frac{M}{2}\right)!}{N_2!\left(\frac{M}{2} - N_2\right)!}.$$



Now using Stirling's approximation again yields:

$$\begin{aligned}\frac{S_1}{k_B} + \frac{S_2}{k_B} \approx \frac{M}{2}\ln\left(\frac{M}{2}\right) - N_1\ln(N_1) - \left(\frac{M}{2} - N_1\right)\ln\left(\frac{M}{2} - N_1\right) + \frac{M}{2}\ln\left(\frac{M}{2}\right) \\ - (N - N_1)\ln(N - N_1) - \left(\frac{M}{2} - (N - N_1)\right)\ln\left(\frac{M}{2} - (N - N_1)\right).\end{aligned}$$

Let us now look for the maximum entropy:

$$\begin{aligned}0 = \frac{1}{k_B}\frac{\partial S}{\partial N_1} &= \frac{1}{k_B}\left(\frac{\partial S_1}{\partial N_1} + \frac{\partial S_2}{\partial N_1}\right) \\ &= -1 - \ln(N_1) + 1 + \ln\left(\frac{M}{2} - N_1\right) + 1 + \ln(N - N_1) - 1 - \ln\left(\frac{M}{2} - (N - N_1)\right) \\ &= \ln\left[\frac{(N - N_1)\left(\frac{M}{2} - N_1\right)}{N_1\left(\frac{M}{2} - (N - N_1)\right)}\right].\end{aligned}$$

Rearranging this expression and solving for N_1 yields $N_1 = \frac{N}{2}$. Hence we conclude that equilibrium corresponds to equal partitioning of atoms. Let us now look at the associated entropy:

$$\begin{aligned} 2S\left(\frac{M}{2},\frac{N}{2}\right) &= 2\left(\frac{M}{2}\ln\left(\frac{M}{2}\right)-\frac{N}{2}\ln\left(\frac{N}{2}\right)-\left(\frac{M}{2}-\frac{N}{2}\right)\ln\left(\frac{M}{2}-\frac{N}{2}\right)\right) \\ &= M(\ln(M)-\ln(2))-N(\ln(N)-\ln(2))-(M-N)(\ln(M-N)-\ln(2)) \\ &= M\ln(M)-N\ln(N)-(M-N)\ln(M-N) \\ &= S(M,N). \end{aligned}$$

However, observe that $\left[\Omega(0,3,2)\right]^2 = \left(\frac{3!}{2!1!}\right)^2 = 9$ *and* $\Omega(0,6,4) = \frac{6!}{4!2!} = 15 \neq 9 = \left[\Omega(0,3,2)\right]^2$. *Hence in this case* $\Omega \neq \Omega_1\Omega_2$, *why is this happening? Atoms can distribute unevenly between the two subsystems, which is untrue if we set* $N_1 = N_2 = \frac{N}{2}$.

- *For nanoscale systems entropy can be non-additive.*
- *This can be seen as a border effect: a system with twice the size is not the same as two independent systems.*

2.2 Statistical Properties of some Ensembles

In this section, we present explicit ensembles and describe the systems with a statistical point of view. We will first give a small introduction through the study of the canonical ensemble before formally introducing the so-called partition function. We will then show some properties of this function and find different ways to calculate thermodynamic quantities with it.

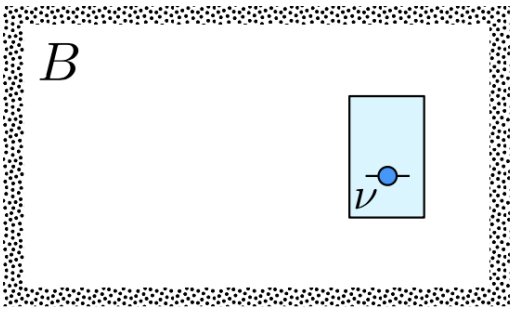
2.2.1 The Partition Function

Let us provide intuition on the definition of the partition function, and to this end consider the canonical ensemble (i.e. constant temperature). Imagine a system coupled to a bath and assume we know the density of states for a large bath $\Omega_B(E)$. The {system + bath} system is kept at constant energy E and the subsystem is in a defined state with energy E_ν. Now the number of states of the bath is going to be $\Omega_B(E-E_\nu)$. We can now expand $\ln(\Omega_B)$ as a Taylor series:

$$\ln(\Omega_B(E-E_\nu)) = \ln(\Omega_B(E)) - E_\nu\frac{\partial\ln(\Omega_B(E))}{\partial E} + ... \tag{2.11}$$

This gives us the probability of a state given its energy $\left(\text{recall } \frac{1}{T} = \frac{\partial S}{\partial E} = k_B\frac{\partial\ln(\Omega)}{\partial E}\right)$:

$$\mathbb{P}_\nu(E_\nu) \propto \Omega_B(E-E_\nu) \propto \exp\left(-E_\nu\frac{\partial\ln(\Omega_B(E))}{\partial E}\right) = \exp\left(-\frac{E_\nu}{k_BT}\right). \tag{2.12}$$



Note that $\mathbb{P}_\nu$ is computed for a single state. If degenerate states with the same energy are present, they have to be counted separately. We would then have:

$$\mathbb{P}(E) \propto \Omega(E) \exp\left(-\frac{E}{k_B T}\right). \tag{2.13}$$

Definition 2.2.1. *We define the* ***partition function*** Q *as the normalization constant in the following formula:*

$$\mathbb{P}_\nu = \frac{e^{-\frac{E_\nu}{k_B T}}}{Q} := \frac{e^{-\frac{E_\nu}{k_B T}}}{\sum_{\nu'} e^{-\frac{E_{\nu'}}{k_B T}}}. \tag{2.14}$$

Classically, the probability involves a density and so Q is an integral over a phase space. In fact, the partition function is the Laplace transform of the density of states Ω, so it has the same content of information $Q(N,V,\beta) = \int \Omega(E,N,V) e^{-\beta E} dE$. Therefore, the partition function is defined as:

$$Q(N,V,T) = \int e^{-\frac{E(\mathbf{q},\mathbf{p})}{k_B T}} d\mathbf{p} d\mathbf{q}. \tag{2.15}$$

For discrete states, Q is a sum over states. In quantum mechanics there is no need to know the energy eigenstates since $Q = \sum_\nu \langle \nu | e^{-\beta \widehat{H}} | \nu \rangle = Tr\left(e^{-\beta \widehat{H}}\right)$ can be evaluated over any complete basis. Hence in this case, the partition function is defined as:

$$Q(N,V,T) = \sum_\nu e^{-\frac{E_\nu}{k_B T}}. \tag{2.16}$$

Definition 2.2.2. *We define the* ***heat capacity*** C_v *as* $C_v := \frac{\partial E}{\partial T} \ \left(= \frac{\partial \langle E \rangle}{\partial T}\right)$.

Remark 2.2.3. *Remark that* $C_v \neq \left(\frac{\partial E}{\partial T}\right)_p$*, where brackets emphasize that the derivative is taken at constant pressure. In other words, we can only get the heat capacity by differentiating the energy for certain variable combinations.*

Proposition 2.2.4. *The internal energy can be obtained from the partition function via:*

$$\langle E \rangle = -\frac{\partial \ln(Q)}{\partial \beta}. \tag{2.17}$$

Proof. With a simple computation we find:

$$-\frac{\partial \ln(Q)}{\partial \beta} = -\frac{1}{Q}\frac{\partial Q}{\partial \beta} = \sum_\nu E_\nu \frac{e^{-\beta E_\nu}}{Q} = \langle E \rangle. \tag{2.18}$$

□

Proposition 2.2.5. *The energy fluctuations can be obtained from the internal energy (or partition function) via:*

$$\mathrm{Var}(E) = -\frac{\partial \langle E \rangle}{\partial \beta} = \frac{\partial^2 \ln(Q)}{\partial \beta^2}. \tag{2.19}$$

Proof. With a simple computation we find:

$$\begin{aligned} -\frac{\partial \langle E \rangle}{\partial \beta} = \frac{\partial^2 \ln(Q)}{\partial \beta^2} = -\frac{\partial}{\partial \beta} \sum_\nu E_\nu \frac{e^{-\beta E_\nu}}{Q} &= \sum_\nu E_\nu^2 \frac{e^{-\beta E_\nu}}{Q} + \sum_\nu E_\nu \frac{e^{-\beta E_\nu}}{Q^2} \frac{\partial Q}{\partial \beta} \\ &= \sum_\nu E_\nu^2 \frac{e^{-\beta E_\nu}}{Q} + \frac{1}{Q} \frac{\partial Q}{\partial \beta} \sum_\nu E_\nu \frac{e^{-\beta E_\nu}}{Q} \\ &= \left\langle E^2 \right\rangle - \langle E \rangle^2 = \mathrm{Var}(E). \end{aligned}$$

□

Finally remark that the fluctuations become negligible in the thermodynamic limit, where C_v denotes the heat capacity:

$$-\frac{\partial \langle E \rangle}{\partial \beta} = -\frac{\partial \langle E \rangle}{\partial T} \frac{\partial T}{\partial \beta} = C_v T^2 k_B \sim \mathcal{O}(N). \tag{2.20}$$

Hence the coefficient of variation behaves as:

$$\frac{\sigma(E)}{\langle E \rangle} \sim \frac{\sqrt{\mathcal{O}(N)}}{\mathcal{O}(N)} = \frac{1}{\sqrt{\mathcal{O}(N)}} \xrightarrow{N \to \infty} 0. \tag{2.21}$$

Example 2.2.6 (Ideal Gas)**.** *Let us find the partition function and the heat capacity for an ideal gas with N atoms in a volume V. Remark that in the following expression $\mathbf{q}$ will yield V, the vectors $\{\mathbf{p}, \mathbf{q}\}$ are both N-dimensional and to avoid counting indistinguishable states we have to consider:*

$$\begin{aligned} Q = \frac{1}{N!} \int \exp\left(-\beta \frac{\mathbf{p}^2}{2m}\right) d\mathbf{p}^N d\mathbf{q}^N = \frac{1}{N!} \int \exp\left(-\frac{\beta}{2m} \sum_i \mathbf{p}_i^2\right) d\mathbf{p}^N d\mathbf{q}^N &= \frac{1}{N!} \left[\left(\frac{m\pi}{\beta}\right)^{\frac{3}{2}} \left(2\sqrt{2}V\right) \right]^N \\ &= \frac{V^N}{N!} \left(2mk_B T\pi\right)^{\frac{3}{2}N}, \end{aligned}$$

where we have used the general formula for a Gaussian integral $\int e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$. Now, with a simple calculation we have that:

$$\ln(Q) = N\left[1 + \frac{3}{2}\ln\left(\frac{m\pi}{\beta}\right) + \ln(V) + \frac{1}{2}\ln(8)\right] - N\ln(N). \tag{2.22}$$

Hence, the average energy is given by:

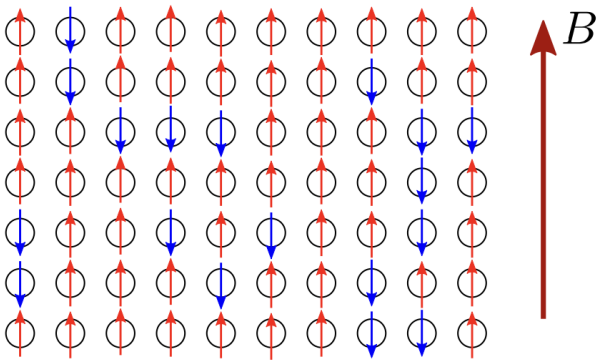
$$\langle E \rangle = -\frac{\partial}{\partial \beta} \ln(Q) = \frac{3N}{2\beta} = \frac{3N}{2} k_B T. \tag{2.23}$$

Finally the heat capacity is simply $C_V = \frac{3N}{2} k_B$.

Example 2.2.7 (Two-State System)**.** *Let us find the partition function and the heat capacity for a two-state quantum system that can exit in two states with energies 0 and ϵ. The partition function is simply $Q = e^0 + e^{-\beta\epsilon} = 1 + e^{-\beta\epsilon}$, and the average energy is given by $\langle E\rangle = -\frac{\partial}{\partial\beta}\ln(Q) = \frac{\epsilon}{e^{\beta\epsilon}+1}$. Finally, in the thermodynamic limits we have: $\langle E\rangle \xrightarrow{T\to 0} 0$ and $\langle E\rangle \xrightarrow{T\to\infty} \frac{\epsilon}{2}$.*

2.2.2 Example - A Paramagnetic Material

Consider a large number N of independent spins in a magnetic field. Each can have two energy levels, $\epsilon = \mu B$ and $\epsilon = -\mu B$. We now make two analyses - one on the microcanonical case (NVE model, i.e. constant N,V,E) and another on the canonical case (NVT model, i.e. constant N,V,T).



• In this first point, we consider non-interacting spins in the microcanonical case. Let us study the density of states. Given m spins in the up state $\uparrow$ and setting $x = \frac{E}{N\epsilon}$, the internal energy is:

$$E = -m\epsilon + (N-m)\epsilon = (N-2m)\epsilon \Longrightarrow m = \frac{1}{2}\left(N - \frac{E}{\epsilon}\right) = \frac{N}{2}(1-x). \tag{2.24}$$

Now the number of ways to arrange m spins in the up state $\uparrow$ and $(N-m)$ spins down $\downarrow$ is:

$$\Omega = \binom{N}{m} = \frac{N!}{m!(N-m)!} \approx e^{N\ln(N) - m\ln(m) - (N-m)\ln(N-m)}, \tag{2.25}$$

where we have used Stirling's Formula for an approximation. Hence, substituting the explicit value of m yields the density of states:

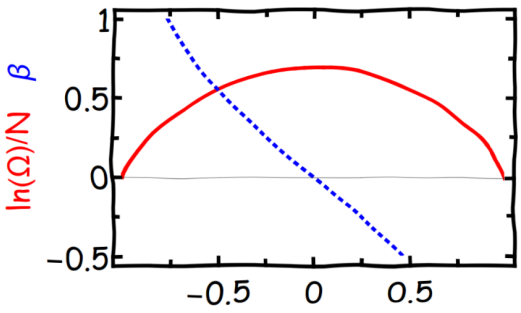
$$\ln(\Omega(E)) = N(\ln(N) + \ln(2)) - \frac{1}{2}\left[\left(N - \frac{E}{\epsilon}\right)\ln\left(N - \frac{E}{\epsilon}\right) + \left(N + \frac{E}{\epsilon}\right)\ln\left(N + \frac{E}{\epsilon}\right)\right]. \tag{2.26}$$

Let us finally look at the inverse temperature for a given energy/number of up spins $\uparrow$:

$$\beta = \frac{\partial\ln(\Omega)}{\partial E} = -\frac{1}{2\epsilon}\left[-1 - \ln\left(N - \frac{E}{\epsilon}\right) + 1 + \ln\left(N + \frac{E}{\epsilon}\right)\right] = \frac{1}{2\epsilon}\ln\left(\frac{1 - \frac{E}{N\epsilon}}{1 + \frac{E}{N\epsilon}}\right). \tag{2.27}$$

Remark that we can invert the relation to obtain E as a function of β:

$$\frac{1 - \frac{E}{N\epsilon}}{1 + \frac{E}{N\epsilon}} = e^{2\beta\epsilon} \Longrightarrow \frac{E}{N\epsilon} = x = \frac{1 - e^{2\beta\epsilon}}{1 + e^{2\beta\epsilon}} = \frac{e^{-\beta\epsilon} - e^{\beta\epsilon}}{e^{\beta\epsilon} + e^{-\beta\epsilon}}. \tag{2.28}$$



Above is a plot of β (in blue) versus $\frac{\ln(\Omega)}{N}$ (in red), and one observes that we can attain negative temperatures.

• In this second point, we consider non-interacting spins in the canonical case. Let us study the partition function. The canonical partition function can be written as a sum over energy levels:

$$Q = \sum_{m=0}^{N} \binom{N}{m} e^{-\beta(N-2m)\epsilon} = \sum_{m=0}^{N} \frac{N!}{m!(N-m)!} e^{-\beta(N-2m)\epsilon} = \sum_{E} e^{-\beta E}\Omega(E). \tag{2.29}$$

Note that in general, given two non-interacting systems, the overall partition function is the product of the two partition functions:

$$Q_1 = \sum_{\nu_1} e^{-\beta E_{\nu_1}}, \quad Q_2 = \sum_{\nu_2} e^{-\beta E_{\nu_2}} \Longrightarrow Q = \sum_{\nu_1,\nu_2} e^{-\beta(E_{\nu_1}+E_{\nu_2})} = Q_1 Q_2. \tag{2.30}$$

Now, each of the N spins can exist in two states $Q_1 = e^{-\beta\epsilon} + e^{\beta\epsilon}$. Hence: $Q = Q_1^N$.
Remark that $E(\beta)$ matches the microcanonical case:

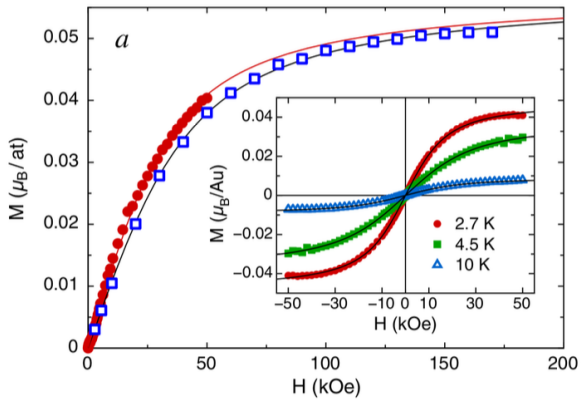
$$\langle E\rangle = -\frac{\partial \ln(Q)}{\partial\beta} = -N\frac{\partial \ln(Q_1)}{\partial\beta} = N\epsilon\frac{e^{-\beta\epsilon} - e^{\beta\epsilon}}{e^{\beta\epsilon} + e^{-\beta\epsilon}}. \tag{2.31}$$

• We end this example by looking at the magnetization. For a system with m spins up $\uparrow$, the magnetization is $M = \mu m - \mu(N-m) = -\frac{\mu}{\epsilon}E$. Hence:

$$\frac{\langle M\rangle}{N} = -\mu\frac{e^{-\beta\epsilon} - e^{\beta\epsilon}}{e^{\beta\epsilon} + e^{-\beta\epsilon}} - \mu\tanh(-\beta\epsilon) = \mu\tanh(\beta\mu B). \tag{2.32}$$

And in fact, for small values of the applied field, one obtains **Curie's Law**:

$$\frac{M}{N} \approx \frac{\mu^2 B}{k_B T}. \tag{2.33}$$



Paramagnetism of gold nanoparticles (Bartolomé et al., PRL 2012)

2.3 The Gibbs Entropy

In this section we introduce the Gibbs entropy formula and present Shannon's context for the information entropy. Until now, we have defined statistical properties assuming an equiprobability criterion. Let us now find a way to define entropy when there is no equiprobability between microstates.

Definition 2.3.1. *We define the* ***Gibbs entropy*** *with the following formula:*

$$\mathcal{S} = -k_B \sum_{\nu} \mathbb{P}_\nu \ln(\mathbb{P}_\nu). \tag{2.34}$$

• Considering the canonical ensemble and given the relation between Q and $\langle E \rangle$: $-\frac{\partial \ln(Q)}{\partial \beta} = \langle E \rangle$ we note the differential form $d\ln(Q) = -\langle E \rangle d\beta$. Now, explicitly using the fact that $\mathbb{P}_\nu = \frac{e^{-\beta E_\nu}}{Q}$, Gibbs entropy formula yields:

$$\frac{\mathcal{S}}{k_B} = -\sum_{\nu} \frac{e^{-\beta E_\nu}}{Q} \ln\left(\frac{e^{-\beta E_\nu}}{Q}\right) = \sum_{\nu} \frac{e^{-\beta E_\nu}}{Q} (\beta E_\nu + \ln(Q)) = \ln(Q) + \beta \langle E \rangle. \tag{2.35}$$

Hence the differential form gives:

$$\frac{d\mathcal{S}}{k_B} = \underbrace{d\ln(Q) + \langle E \rangle d\beta}_{=0} + \beta d\langle E \rangle = \beta d\langle E \rangle. \tag{2.36}$$

This result is consistent with the classical thermodynamic definition of entropy for a system whose is a reversible constant-(T,V) transformation:

$$dS = \frac{d\langle E \rangle}{T} = k_B \beta d\langle E \rangle. \tag{2.37}$$

• Gibbs entropy has a deep connection with information theory. Given an event A that occurs with probability $\mathbb{P}(A)$, let us define the *information content* $I(A)$ of that event. For this we require: (i) $I(A) \geq 0$, (ii) $I(\Omega) = 0$ (an event that always occurs does not convey information), and (iii) if A and B are independent events then $I(A, B) = I(A) + I(B)$. The only function of $\mathbb{P}(A)$ that satisfies these conditions is:

$$I(A) = C \log_b\left(\frac{1}{\mathbb{P}(A)}\right). \tag{2.38}$$

Given a set of disjoint events A_i, the average information that we can obtain equals the information of all events weighed with their probability. We now have Shannon's framework:

Definition 2.3.2. *We define the* ***Shannon information entropy*** *with the following formula:*

$$I = -\sum_{i} \mathbb{P}(A_i) \ln\left(\mathbb{P}(A_i)\right). \tag{2.39}$$

Proposition 2.3.3 (Additivity)**.** *Considering two non-interacting subsystems A and B:*

$$\frac{\mathcal{S}}{k_B} = \frac{\mathcal{S}_A}{k_B} + \frac{\mathcal{S}_B}{k_B}. \tag{2.40}$$

Proof. Set two non-interacting subsystems A and B:

$$\frac{\mathcal{S}_A}{k_B} = -\sum_{\nu_A} \mathbb{P}_A(\nu_A) \ln\big(\mathbb{P}_A(\nu_A)\big) \quad \text{and,} \quad \frac{\mathcal{S}_B}{k_B} = -\sum_{\nu_B} \mathbb{P}_B(\nu_B) \ln\big(\mathbb{P}_B(\nu_B)\big). \tag{2.41}$$

Now, if we consider them as a single system, possible states will include all possible pairs (ν_A, ν_B) occurring with probabilities $\mathbb{P}(\nu_A, \nu_B) = \mathbb{P}_A(\nu_A)\mathbb{P}_B(\nu_B)$. Hence considering the overall Gibbs entropy we obtain:

$$\begin{aligned}
-\frac{\mathcal{S}}{k_B} &= \sum_{(\nu_A,\nu_B)} \mathbb{P}(\nu_A,\nu_B) \ln\big(\mathbb{P}(\nu_A,\nu_B)\big) \\
&= \sum_{(\nu_A,\nu_B)} \mathbb{P}_A(\nu_A)\mathbb{P}_B(\nu_B)\Big(\ln\big(\mathbb{P}_A(\nu_A)\big) + \ln\big(\mathbb{P}_B(\nu_B)\big)\Big) \\
&= \sum_{(\nu_A,\nu_B)} \mathbb{P}_A(\nu_A)\mathbb{P}_B(\nu_B) \ln\big(\mathbb{P}_A(\nu_A)\big) + \sum_{(\nu_A,\nu_B)} \mathbb{P}_A(\nu_A)\mathbb{P}_B(\nu_B) \ln\big(\mathbb{P}_B(\nu_B)\big) \\
&= -\frac{\mathcal{S}_A}{k_B} - \frac{\mathcal{S}_B}{k_B},
\end{aligned}$$

where in the last step we have used the fact that $\sum_\nu \mathbb{P}_\nu = 1$.

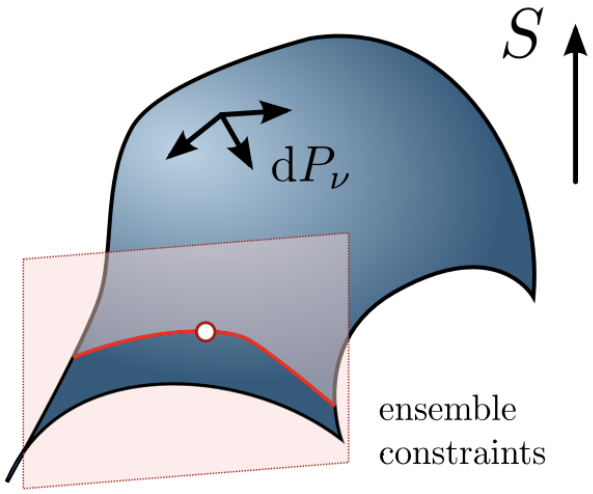
□

2.4 Some Ensembles in the Context of MaxEnt

In this section, we study several examples of ensembles using a maximum entropy (MaxEnt) study. The general method will make use of Lagrange multipliers and applying the appropriate constraints on the system. For the trick, we infer an expression for $\mathbb{P}_\nu$ by imposing the equilibrium condition $\delta\mathcal{S} = 0$ and also imposing the conservation conditions characteristic of the ensemble subject to normalization $\sum_\nu \mathbb{P}_\nu = 1$:

$$\langle E \rangle = \sum_\nu E_\nu \mathbb{P}_\nu, \qquad \langle N \rangle = \sum_\nu N_\nu \mathbb{P}_\nu. \tag{2.42}$$

We will see that maximizing entropy means choosing the probability distribution of microstates that encodes the least amount of information. However, we will do so consistently with the thermodynamic constraints imposed by the ensemble.



2.4.1 The Microcanonical Ensemble

Let us start with the microcanonical ensemble (constant NVE ensemble), where $E_\nu = E$, $N_\nu = N$ and the states are strictly restricted to those fulfilling these conditions. We can now use a Lagrange multiplier γ to enforce the normalization condition in $\delta\mathcal{L}$:

$$\delta\left(S+\gamma\sum_\nu \mathbb{P}_\nu\right)=0. \tag{2.43}$$

Expanding the differential form and using the chain rule on $S=-k_B\sum_\nu \mathbb{P}_\nu \ln\left(\mathbb{P}_\nu\right)$ we obtain:

$$0=\sum_\nu\left(\frac{\partial S}{\partial \mathbb{P}_\nu}d\mathbb{P}_\nu+\gamma d\mathbb{P}_\nu\right)=\sum_\nu\left(-k_B(1+\ln(\mathbb{P}_\nu))+\gamma\right)d\mathbb{P}_\nu. \tag{2.44}$$

The only way the differential form can be zero for any choice of $d\mathbb{P}_\nu$ is if the coefficient is zero. Hence:

$$\ln(\mathbb{P}_\nu)=\frac{\gamma}{k_B}-1\Longrightarrow \mathbb{P}_\nu=\text{constant}=\frac{\delta(E-E_\nu)}{\Omega(E)}. \tag{2.45}$$

2.4.2 The Canonical Ensemble

We now consider the constant NVT ensemble. In this case, we also want to enforce a proper value of the mean energy $\langle E\rangle$, hence we need to introduce a new Lagrange multiplier. We consider $\delta\mathcal{L}$ as:

$$\delta\left(S+\gamma\sum_\nu \mathbb{P}_\nu+\alpha\sum_\nu E_\nu\mathbb{P}_\nu\right)=0. \tag{2.46}$$

Again expanding the differential form and using the chain rule yields:

$$0=\sum_\nu\left(\frac{\partial S}{\partial \mathbb{P}_\nu}d\mathbb{P}_\nu+\gamma d\mathbb{P}_\nu+\alpha E_\nu d\mathbb{P}_\nu\right)=\sum_\nu\left(-k_B(1+\ln(\mathbb{P}_\nu))+\gamma+\alpha E_\nu\right)d\mathbb{P}_\nu. \tag{2.47}$$

This leads up to:

$$-k_B(1+\ln(\mathbb{P}_\nu))+\gamma+\alpha E_\nu=0\Longrightarrow \ln(\mathbb{P}_\nu)=\frac{\gamma+\alpha E_\nu-k_B}{k_B}. \tag{2.48}$$

One of the two multipliers is set by the normalization. For the other, we need a constraint, e.g. the condition $\beta=\frac{1}{k_B}\left(\frac{\partial S}{\partial E}\right)_{V,N}$. Exploiting the fixed normalization that ensures $\sum_\nu d\mathbb{P}_\nu=0$ yields:

$$\frac{dS}{k_B}=-\sum_\nu(1+\ln(\mathbb{P}_\nu))d\mathbb{P}_\nu=-\frac{1}{k_B}\sum_\nu(\gamma+\alpha E_\nu)d\mathbb{P}_\nu=-\frac{\alpha}{k_B}d\langle E\rangle. \tag{2.49}$$

Therefore, we have the following series of implications:

$$\alpha=-\left(\frac{\partial S}{\partial E}\right)_{V,N}=-k_B\beta\Longrightarrow \mathbb{P}_\nu=e^{-\beta E_\nu}e^{\frac{\gamma}{k_B}-1}. \tag{2.50}$$

• The connection between the Lagrange multiplier α and the inverse temperature β elucidates the relation between the statistical mechanical formulation and several thermodynamic variables. Recall

(2.48) and (2.50) such that $\alpha = -k_B\beta = -\frac{1}{T}$, then the Gibbs entropy reads:

$$\mathcal{S} = -k_B \sum_\nu \mathbb{P}_\nu \ln(\mathbb{P}_\nu) = \sum_\nu (k_B - \gamma + k_B\beta E_\nu)\mathbb{P}_\nu = k_B\beta\langle E\rangle + k_B - \gamma. \tag{2.51}$$

In particular, the free energy (in this course $\langle A\rangle = A$ and $\langle E\rangle = E$) is:

$$\langle E\rangle - T\mathcal{S} = \gamma T - k_B T = \langle A\rangle. \tag{2.52}$$

Hence the probability distribution reads:

$$\mathbb{P}_\nu = e^{-\beta E_\nu} e^{\beta(\gamma T - k_B T)} = e^{-\beta(E_\nu - \langle A\rangle)}. \tag{2.53}$$

Finally considering that $Q = \sum_\nu e^{-\beta E_\nu}$ and that $1 = \sum_\nu \mathbb{P}_\nu = e^{\beta\langle A\rangle}\sum_\nu e^{-\beta E_\nu}$ we have:

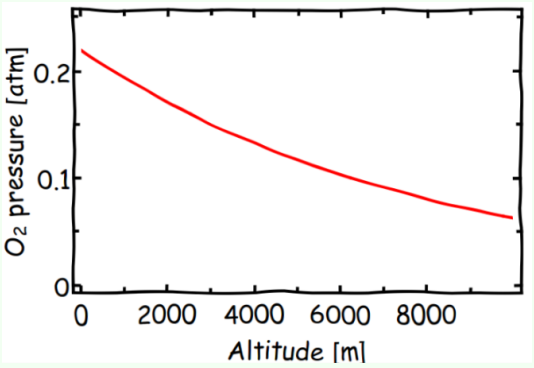
$$e^{\beta\langle A\rangle} = \frac{1}{Q} \Longrightarrow \langle A\rangle = -k_B T \ln(Q). \tag{2.54}$$

Example 2.4.1 (Particles in an External Potential)**.** *Consider a system in which the energy of individual particles depends on position. For instance, for a gas in a uniform gravitational field $E(h) = mgh$. For each particle there is a uniform density of states along h, and the probability of finding a particle at an altitude h reads:*

$$\mathbb{P}(h) = \frac{1}{Q} e^{-\beta mgh} = \frac{e^{-\beta mgh}}{\beta g m}. \tag{2.55}$$

The density as a function of altitude is then proportional to $\mathbb{P}(h)$. Concentration of gases vary with altitude:

$$\rho(h) = \rho_0 e^{-\beta mgh} \quad \textit{and in general,} \quad \rho_i(h) = \frac{\rho_0^i e^{-\beta m_i gh}}{\sum_j \rho_0^j e^{-\beta m_j gh}}. \tag{2.56}$$



2.4.3 The Grand-Canonical Ensemble

We now consider the constant chemical potential μ case (constant μVT ensemble). However, we leave the number of particles N and the energy E free to fluctuate. We now need three Lagrange multipliers for $\delta\mathcal{L}$:

$$\delta\left(S + \gamma\sum_\nu \mathbb{P}_\nu + \alpha\sum_\nu E_\nu\mathbb{P}_\nu + \eta\sum_\nu N_\nu\mathbb{P}_\nu\right) = 0. \tag{2.57}$$

Writing the differential form explicitly gives an expression for $\mathbb{P}_\nu$:

$$\sum_\nu \big(-k_B(1+\ln(\mathbb{P}_\nu))+\gamma+\alpha E_\nu+\eta N_\nu\big)d\mathbb{P}_\nu = 0 \Longrightarrow \ln(\mathbb{P}_\nu) = \frac{\gamma - k_B + \alpha E_\nu + \eta N_\nu}{k_B}. \tag{2.58}$$

We can now use this to work out an expression for $d\mathcal{S}$:

$$d\mathcal{S} = -\sum_\nu k_B(1+\ln(\mathbb{P}_\nu))d\mathbb{P}_\nu = -\sum_\nu(\alpha E_\nu+\eta N_\nu)d\mathbb{P}_\nu = -\alpha d\langle E\rangle - \eta d\langle N\rangle. \tag{2.59}$$

Now given that $\left(\frac{\partial \mathcal{S}}{\partial N}\right)_{E,V} = -\left(\frac{\partial E}{\partial N}\right)_{V,S}/\left(\frac{\partial E}{\partial S}\right)_{V,N} = -\frac{\mu}{T}$, macroscopic thermodynamic relations fix the values of α and η:

$$\left(\frac{\partial \mathcal{S}}{\partial E}\right)_N = -\alpha \Longrightarrow \alpha = -\frac{1}{T} = -k_B\beta, \tag{2.60}$$

and,

$$\left(\frac{\partial \mathcal{S}}{\partial N}\right)_E = -\eta \Longrightarrow \eta = \frac{\mu}{T} = k_B\beta\mu. \tag{2.61}$$

With these expressions (2.60) and (2.61) we can write out the probability distribution. The remaining multiplier γ can be determined from normalization:

$$\ln(\mathbb{P}_\nu) = \frac{\gamma}{k_B} - 1 - \beta E_\nu + \beta\mu N_\nu. \tag{2.62}$$

We can however relate γ to macroscopic thermodynamics. First write out the entropy:

$$\mathcal{S} = -k_B\sum_\nu \mathbb{P}_\nu \ln(\mathbb{P}_\nu) = k_B - \gamma + \frac{\langle E\rangle}{T} - \frac{\mu}{T}\langle N\rangle. \tag{2.63}$$

Recall that $\langle G\rangle = \mu\langle N\rangle = \langle E\rangle - TS + \rho V$. We can then write out an expression for $(\gamma - k_B)$ based on the Gibbs free energy:

$$T(\gamma - k_B) = \langle E\rangle - TS - \mu\langle N\rangle = \langle E\rangle - TS - \big(\langle E\rangle - TS + \rho V\big) = -\rho V. \tag{2.64}$$

So the normalized $\mathbb{P}_\nu$ reads:

$$\mathbb{P}_\nu = \exp\big(-\beta(E_\nu - \mu N_\nu + \rho V)\big). \tag{2.65}$$

We can, as a result, introduce the grand-canonical partition function Ξ:

$$\Xi = \sum_\nu \exp\big(-\beta(E_\nu - \mu N_\nu)\big) = e^{\beta\rho V} \Longrightarrow \langle \rho V\rangle = \frac{1}{\beta}\ln(\Xi). \tag{2.66}$$

2.4.4 Isothermal-Isobaric Ensemble

Let us present a last example of an ensemble where V and E can fluctuate by coupling the system to a bath:

$$\delta\left(S + \gamma\sum_\nu \mathbb{P}_\nu + \alpha\sum_\nu E_\nu\mathbb{P}_\nu + \pi\sum_\nu V_\nu\mathbb{P}_\nu\right) = 0. \tag{2.67}$$

With the same calculations as in the previous subsection we have:

$$\ln(\mathbb{P}_\nu) = \frac{\gamma - k_B + \alpha E_\nu + \pi V_\nu}{k_B}. \tag{2.68}$$

As usual, considering $\left(\frac{\partial \mathcal{S}}{\partial V}\right)_E = -\left(\frac{\partial E}{\partial V}\right)_S / \left(\frac{\partial E}{\partial S}\right)_V = \frac{\rho}{T}$, we may write the differential of $\mathcal{S}$:

$$d\mathcal{S} = -\sum_\nu (\alpha E_\nu + \pi V_\nu) d\mathbb{P}_\nu = -\alpha d\langle E\rangle - \pi d\langle V\rangle \Longrightarrow \alpha = -\frac{1}{T} \quad \text{and,} \quad \pi = -\frac{\rho}{T}. \tag{2.69}$$

Hence we can write:

$$\mathbb{P}_\nu = \Delta^{-1} e^{-\beta(E_\nu + \rho V_\nu)} \quad \text{and,} \quad G = -k_B T \ln(\Delta). \tag{2.70}$$

2.5 System Fluctuations

This section is dedicated to a changing number of particles N in the system.

2.5.1 Particle Number Fluctuations

Let us consider the case of two components with particle numbers N_ν^A and N_ν^B. As usual, and similarly to the grand-canonical ensemble, set $\delta\mathcal{L}$ to be:

$$\delta\left(S + \gamma \sum_\nu \mathbb{P}_\nu + \alpha \sum_\nu E_\nu \mathbb{P}_\nu + \eta^A \sum_\nu N_\nu^A \mathbb{P}_\nu + \eta^B \sum_\nu N_\nu^B \mathbb{P}_\nu\right) = 0. \tag{2.71}$$

Then we obtain:

$$\ln(\mathbb{P}_\nu) = \frac{\gamma - k_B + \alpha E_\nu + \eta^A N_\nu^A + \eta^B N_\nu^B}{k_B}. \tag{2.72}$$

And so:

$$\mathbb{P}_\nu = \Xi^{-1} \exp\Big(-\beta(E_\nu - \mu^A N_\nu^A - \mu^B N_\nu^B)\Big). \tag{2.73}$$

The fluctuations in the number of particles and compositions vanish in the thermodynamic limit. For the mean:

$$\langle N^A\rangle = \sum_\nu N_\nu^A \mathbb{P}_\nu = \frac{\partial \ln(\Xi)}{\partial(\beta\mu^A)}. \tag{2.74}$$

Proposition 2.5.1. *For the variance we have:*

$$\mathrm{Var}(N^A) = \frac{\partial^2 \ln(\Xi)}{\partial(\beta\mu^A)^2}. \tag{2.75}$$

Proof. By a direct calculation we find:

$$\frac{\partial^2 \ln(\Xi)}{\partial(\beta\mu^A)^2} = \frac{\partial\langle N^A\rangle}{\partial(\beta\mu^A)} = \sum_\nu N_\nu^A \frac{\partial \mathbb{P}_\nu}{\partial(\beta\mu^A)} = \sum_\nu \left(N_\nu^A\right)^2 \mathbb{P}_\nu - N_\nu^A \mathbb{P}_\nu \frac{1}{\Xi}\frac{\partial \Xi}{\partial(\beta\mu^A)} = \mathrm{Var}(N^A). \tag{2.76}$$

□

Proposition 2.5.2. *For a composition we have:*

$$\langle N^A, N^B\rangle = \frac{\partial^2 \ln(\Xi)}{\partial(\beta\mu^A)\partial(\beta\mu^B)}. \tag{2.77}$$

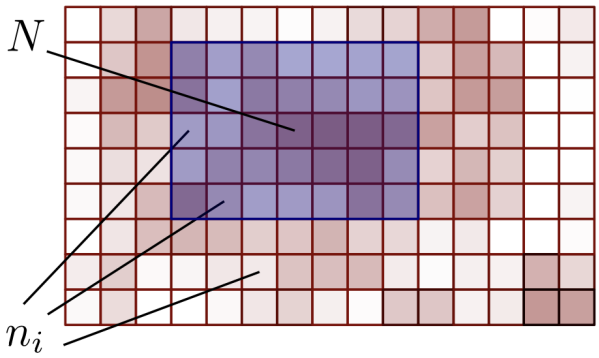
Proof. By a direct calculation we find:

$$\frac{\partial^2 \ln(\Xi)}{\partial(\beta\mu^A)\partial(\beta\mu^B)} = \frac{\partial\langle N^A\rangle}{\partial(\beta\mu^B)} = \sum_\nu N_\nu^A N_\nu^B \mathbb{P}_\nu - N_\nu^A \mathbb{P}_\nu \frac{1}{\Xi}\frac{\partial \Xi}{\partial(\beta\mu^B)} = \langle N^A, N^B\rangle. \tag{2.78}$$

□

2.5.2 Uncorrelated Fluctuations

Let us consider a large reservoir filled with a gas of mean density $\overline{\rho}$. Partition it in disjoint cells of volume V and consider the number of particles that is observed within a region containing M of such cells.



In the above representation, $N = \sum_i n_i$ and $\langle N\rangle = \sum_i \langle n_i\rangle = M\langle n\rangle = MV\overline{\rho}$. Now consider the variance of N with the central assumption of uncorrelated fluctuations $\langle n_i n_j\rangle \overset{i\neq j}{=} \langle n_i\rangle\langle n_j\rangle$.

Proposition 2.5.3. *Under the assumption of uncorrelated fluctuations* $\langle n_i n_j\rangle \overset{i\neq j}{=} \langle n_i\rangle\langle n_j\rangle$*, the variance is:*

$$\mathrm{Var}(N) = M\left(\langle n^2\rangle - \langle n\rangle^2\right) =: M\cdot\mathrm{Var}(n). \tag{2.79}$$

Proof. Directly from the definition of the variance we have:

$$\begin{aligned}
\mathrm{Var}(N) = \left\langle (N - \langle N\rangle)^2\right\rangle = \left\langle \left(\sum_i (n_i - \langle n_i\rangle)\right)^2\right\rangle &= \sum_{i,j}\langle n_i n_j\rangle - M^2\langle n\rangle^2 \\
&= \sum_i \left\langle n_i^2\right\rangle + \sum_{i\neq j}\langle n_i n_j\rangle - M^2\langle n\rangle^2 \\
&= M\left\langle n^2\right\rangle + M(M-1)\langle n\rangle^2 - M^2\langle n\rangle^2 \\
&= M\left(\langle n^2\rangle - \langle n\rangle^2\right) = M\cdot\mathrm{Var}(n).
\end{aligned}$$

□

Remark that relative fluctuations are scale-independent, as using that $\langle N\rangle = M\langle n\rangle$:

$$\mathrm{Var}(N) = M\left(\langle n^2\rangle - \langle n\rangle^2\right) := M\cdot\mathrm{Var}(n) \Longrightarrow \frac{\mathrm{Var}(N)}{\langle N\rangle} = \frac{\mathrm{Var}(n)}{\langle n\rangle}. \tag{2.80}$$

Now, we assume that the volume V of the cells is small enough that the probability of having more than one particle is negligible (alternatively, one can consider that different particles are independent

by considering the probability of having k particles as $(\overline{\rho}V)^k$ and taking the limit $V \to 0$):

$$\langle n \rangle = 1 \cdot \overline{\rho}V + \mathcal{O}(V^2) = \overline{\rho}V + \mathcal{O}(V^2) \quad \text{and,} \quad \langle n^2 \rangle = 1^2 \cdot \overline{\rho}V + \mathcal{O}(V^2) = \overline{\rho}V + \mathcal{O}(V^2) \Longrightarrow \frac{\mathrm{Var}(n)}{\langle n \rangle} \xrightarrow{V \to 0} 1.$$

These considerations will hold as long as fluctuations are uncorrelated on the scale on which we can consider the probability of having more than one particle per cell to be negligible. This is the *dilute limit*.

2.5.3 Ideal Gas Law

In this subsection, we will freely use the Maxwell relation $\left(\frac{\partial(\rho\beta)}{\partial N}\right)_\beta = -\left(\frac{\partial(\beta\mu)}{\partial V}\right)_\beta$, obtained from $\left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T}$ and $\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{p}{T}$. Recall that p is the pressure and $\rho = \frac{N}{V}$ is the density. Now let us combine the results from the last section with the grand-canonical relation and $\frac{\mathrm{Var}(N)}{\langle N \rangle} = \frac{\mathrm{Var}(n)}{\langle n \rangle} = 1$:

$$\left(\frac{\partial \langle N \rangle}{\partial(\beta\mu)}\right)_{\beta,V} = \mathrm{Var}(N) \Longrightarrow \mathrm{Var}(N) = \langle N \rangle. \tag{2.81}$$

We get the relation for the chemical potential in a dilute system:

$$\left(\frac{\partial(\beta\mu)}{\partial \langle N \rangle}\right)_{\beta,V} = \frac{1}{\langle N \rangle} \Longrightarrow \beta\mu = \ln\left(\langle N \rangle\right) + \text{constant}. \tag{2.82}$$

which one can also write as, dividing $\langle N \rangle$ by V:

$$\left(\frac{\partial(\beta\mu)}{\partial \rho}\right)_{\beta} = \frac{1}{\rho} \Longrightarrow \beta\mu = \ln(\rho) + \text{constant}. \tag{2.83}$$

Finally, we use the chain rule to find:

$$\frac{\partial(\beta p)}{\partial \rho} = \frac{\partial \langle N \rangle}{\partial \rho}\frac{\partial(\beta p)}{\partial \langle N \rangle} = -V\frac{\partial(\beta\mu)}{\partial V} = -V\frac{\partial(\beta\mu)}{\partial \rho}\frac{\partial \rho}{\partial V} = \rho\frac{\partial(\beta\mu)}{\partial \rho} = 1. \tag{2.84}$$

Hence, considering that pressure should vanish at zero density, we obtain the **Ideal Gas Law**:

$$\beta p = \rho \quad \text{and,} \quad pV = k_B T \langle N \rangle. \tag{2.85}$$

2.5.4 Example - Langmuir Adsorption Isotherm

Consider the number of ways of arranging M molecules on N surface sites. The energy for M adsorbed molecules with each an energy of ϵ is ϵM. Then:

$$\Omega = \binom{N}{M} = \frac{N!}{M!(N-M)!}. \tag{2.86}$$

Including density of states, the grand-canonical partition function is given by:

$$\mathbb{P}(M) = \Xi^{-1}\frac{N!}{M!(N-M)!}e^{-\beta((\epsilon-\mu)M)} \approx \Xi^{-1}e^{-\beta((\epsilon-\mu)M)+N\ln(N)-M\ln(M)-(N-M)\ln(N-M)}. \tag{2.87}$$

The chemical potential is set to be equal to that in the gas phase (ideal gas limit):

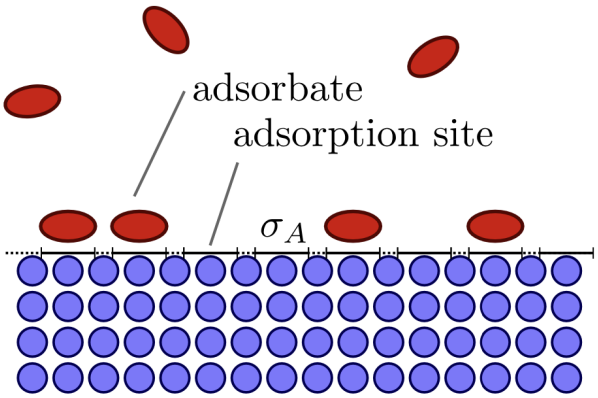
$$e^{\beta\mu} = \frac{\rho}{\rho^{\ominus}} = \frac{p}{p^{\ominus}}. \tag{2.88}$$

Now, setting $\theta = \frac{M}{N}$ to be the *coverage*, if we look for the maximum in $\ln(\mathbb{P})$ we obtain:

$$0 = \frac{\partial \ln(\mathbb{P})}{\partial M} = -\beta\epsilon + \ln\left(\frac{p}{p^{\ominus}}\right) - \ln(M) + \ln(N-M). \tag{2.89}$$

This directly yields:

$$\frac{1}{\theta} = \frac{p^{\ominus}}{p} e^{\beta\epsilon} + 1 \Longrightarrow \theta = \frac{p}{p + p^{\ominus} e^{\beta\epsilon}}. \tag{2.90}$$



2.6 Independent and Indistinguishable Particles (Quantum Approach)

A quantum mechanical (micro)state is characterized in terms of the eigenstates of a complete set of commuting observables. An ensemble can be seen as the set of probabilities of finding the system in one such eigenstate. In this context, the density matrix provides a formally elegant solution to computing ensemble averages in quantum mechanics. In this section we consider the canonical density matrix defined as:

$$\rho = e^{-\beta\widehat{H}}. \tag{2.91}$$

Thus we may consider the eigenvalue/eigenvector equation $\widehat{H}\,|\nu\rangle = E_\nu\,|\nu\rangle$.

Proposition 2.6.1. *The partition function of the density matrix ρ is given by:*

$$Q = \mathrm{Tr}(\rho). \tag{2.92}$$

Proof. This is just a simple calculation from the definition of the trace:

$$\mathrm{Tr}(\rho) = \sum_\nu \langle\nu|\, e^{-\beta\widehat{H}}\,|\nu\rangle = \sum_\nu e^{-\beta E_\nu} = Q. \tag{2.93}$$

□

Proposition 2.6.2. *The expectation of any observable $\widehat{A}$ may be obtained as a trace by:*

$$\left\langle \widehat{A} \right\rangle = \frac{1}{Q} \mathrm{Tr}\left(\rho \widehat{A}\right). \tag{2.94}$$

Proof. This is just a simple calculation from the definition of the trace, along with the eigenstate equation $\widehat{A}\,|\nu\rangle = A_\nu\,|\nu\rangle$ for the observable $\widehat{A}$:

$$\frac{1}{Q} \mathrm{Tr}\left(\rho \widehat{A}\right) = \frac{1}{Q} \sum_\nu \langle \nu |\, e^{-\beta \widehat{H}} \widehat{A}\, |\nu\rangle = \frac{1}{Q} \sum_\nu A_\nu e^{-\beta E_\nu} = \left\langle \widehat{A} \right\rangle. \tag{2.95}$$

□

Let us now see why the Quantum formalism arises by relating a quantum mechanical partition function to a classical canonical partition function. Let us consider the following quantum mechanical partition function in the position representation:

$$Q = \mathrm{Tr}\left(e^{-\beta \widehat{H}}\right) = \int \langle \mathbf{q} |\, e^{-\beta \widehat{H}}\, |\mathbf{q}\rangle \, d\mathbf{q}. \tag{2.96}$$

Let us now pretend that the commutator $\left[\widehat{T}, \widehat{V}\right] = 0$ so we can write $e^{-\beta \widehat{H}} = e^{-\beta \widehat{T}} e^{-\beta \widehat{V}}$:

$$\int \langle \mathbf{q} |\, e^{-\beta \widehat{T}} e^{-\beta \widehat{V}}\, |\mathbf{q}\rangle \, d\mathbf{q} = \int e^{-\beta V(\mathbf{q})} \langle \mathbf{q} |\, e^{-\beta \widehat{T}}\, |\mathbf{q}\rangle \, d\mathbf{q}. \tag{2.97}$$

We now need to insert a $\mathbf{p}$-projector $\int |\mathbf{p}\rangle \langle \mathbf{p}|\, d\mathbf{p}$ and remind ourselves that $\langle \mathbf{p}|\mathbf{q}\rangle \propto e^{i\mathbf{p}\cdot\mathbf{q}}$ leading to $\langle \mathbf{q}|\mathbf{p}\rangle \langle \mathbf{p}|\mathbf{q}\rangle = \text{cst.}$ to find:

$$\int d\mathbf{q} \int e^{-\beta V(\mathbf{q})} \langle \mathbf{q} |\, e^{-\beta \widehat{T}}\, |\mathbf{p}\rangle \langle \mathbf{p}|\mathbf{q}\rangle = \int d\mathbf{q} \int e^{-\beta \left[V(\mathbf{q}) + \frac{p^2}{2m}\right]} \langle \mathbf{q}|\mathbf{p}\rangle \langle \mathbf{p}|\mathbf{q}\rangle \, d\mathbf{p}. \tag{2.98}$$

We are thus left with the classical canonical partition function. Hence a quantum behavior arises from non-commuting operators.

2.6.1 Independent Particles

Let us look at the canonical partition function of two non-interacting subsystems A and B considered as a whole. For this, let ν_i^A and ν_i^B be the microstates of subsystems A and B thus leading to, for $w \in \{A, B\}$:

$$Q^w = \sum_i e^{-\beta E_i^w}. \tag{2.99}$$

Thus considering the composite system, its states correspond to the cartesian product $\{\nu_i^A\} \times \{\nu_i^B\}$ hence giving:

$$Q^{AB} = \sum_{i,j} e^{-\beta E_i^A} e^{-\beta E_j^B} = Q^A Q^B. \tag{2.100}$$

The observables of the composite system are simply:

$$\langle E \rangle = \langle E^A \rangle + \langle E^B \rangle, \tag{2.101}$$

and we obtain uncorrelatedness of individual systems, i.e.:

$$\langle E^A E^B \rangle = \frac{1}{Q} \sum_{i,j} E_i^A E_j^B e^{-\beta\left(E_i^A + E_j^B\right)} = \frac{1}{Q_A Q_B} \left[\sum_i E_i^A e^{-\beta E_i^A} \right] \left[\sum_j E_j^B e^{-\beta E_j^B} \right] = \langle E^A \rangle \langle E^B \rangle. \quad (2.102)$$

Let's now look at states of multi-particles. To this end, consider a system of N particles being described by the many-body wavefunction:

$$\langle x_1, x_2, ..., x_N | \nu \rangle = \Psi_\nu(x_1, x_2, ..., x_N). \quad (2.103)$$

In many cases, we can make a factorization ansatz by simply assuming that particles behave independently of each other:

$$\Psi_\nu(x_1, x_2, ..., x_N) = \prod_{i=1}^{N} \psi_{\nu_i}^{(i)}(x_i). \quad (2.104)$$

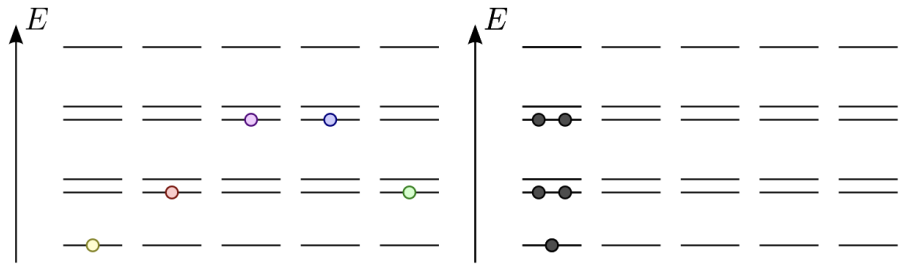
In fact, if all the particles are subject to a sum of single-particle Hamiltonians, then everything reduces to the density matrix and partition function to be the products of single-particle density matrix and partition function. Formally:

$$\rho = e^{-\beta \widehat{H}} = e^{-\beta \sum_i \widehat{H}_i} = \prod_i e^{-\beta \widehat{H}_i} = \prod_i \rho_i. \quad (2.105)$$

$$Q = \mathrm{Tr}\left(e^{-\beta \widehat{H}} \right) = \prod_i \sum_{\nu_i} \langle \nu_i | e^{-\beta \widehat{H}_i} | \nu_i \rangle = \prod_i Q_i. \quad (2.106)$$

2.6.2 Indistinguishability

We now consider what should happen if all the Hamiltonians are identical but the different particles are somehow distinguishable (e.g. they could be far from each other). If particles are all the same then some states are indistinguishable and we must avoid double-counting. An effective method uses occupation numbers for the different levels.



Quantum mechanically, these observables must not change if we swap indices of identical particles. For example we must consider:

$$|\Psi(x_1, x_2, ..., x_N)|^2 = |\Psi(x_2, x_1, ..., x_N)|^2. \quad (2.107)$$

In this case there are only two possibilities. We can either have *bosons* in which case $\Psi(x_1, x_2, ..., x_N) = \Psi(x_2, x_1, ..., x_N)$ and thus:

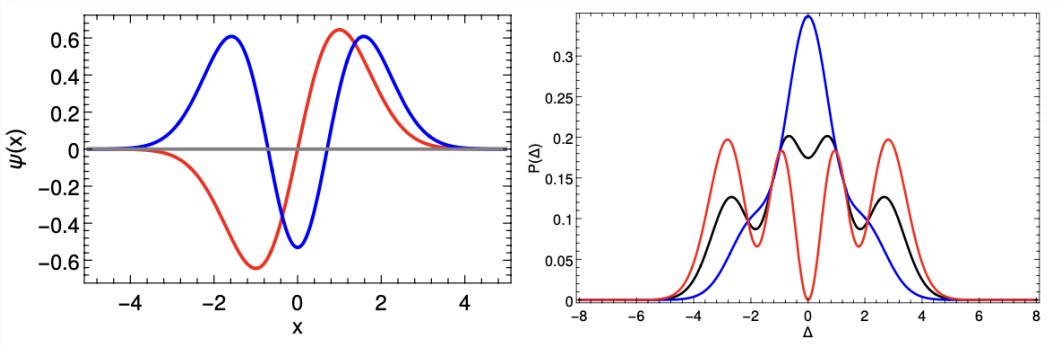
$$\Psi = \frac{1}{\sqrt{N!}} \sum_\pi \prod_{i=1}^{N} \psi_{\nu_i}(x_{\pi_i}), \quad (2.108)$$

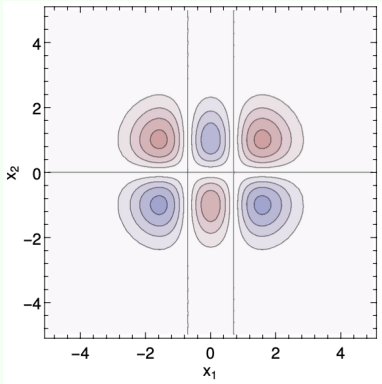
or we may consider *fermions* in which case $\Psi(x_1, x_2, ..., x_N) = -\Psi(x_2, x_1, ..., x_N)$ and thus:

$$\Psi = \frac{1}{\sqrt{N!}} \sum_{\pi} \text{sign}(\pi) \prod_{i=1}^{N} \psi_{\nu_i}(x_{\pi_i}). \tag{2.109}$$

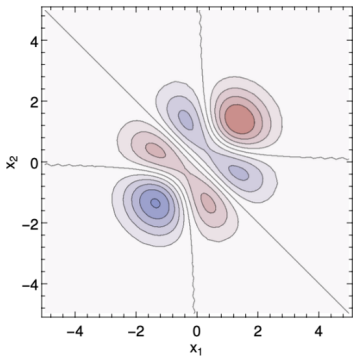
Proposition 2.6.3 (Exclusion Principle)**.** *Ψ_F is zero if two systems are in the same quantum state. Hence each state can only be occupied once.*

Example 2.6.4. *Consider two identical oscillators respectively in the first and second excited states, thus considering two waves $\psi_1(x)$ and $\psi_2(x)$. In this case, the product ansatz $\psi(x_1,x_2) = \psi_1(x_2)\psi_2(x_1)$ is not invariant to particle exchanges and the symmetrized wavefunction $\psi(x_1,x_2) = \frac{1}{\sqrt{2}}[\psi_1(x_1)\psi_2(x_2) \pm \psi_1(x_2)\psi_2(x_1)]$ introduces new correlations. Below are two plots of ψ_1, ψ_2, the probability $\mathbb{P}(\Delta) := \int |\Psi(x, x+\Delta)|^2 dx$ and the two cases for $\psi(x_1,x_2)$.*

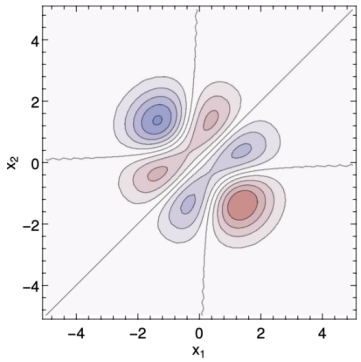




$$\Psi(x_1, x_2) = \psi_1(x_2)\,\psi_2(x_1)$$



$$\Psi(x_1, x_2) = \frac{1}{\sqrt{2}}\left[\psi_1(x_1)\,\psi_2(x_2) + \psi_1(x_2)\,\psi_2(x_1)\right]$$



$$\Psi(x_1, x_2) = \frac{1}{\sqrt{2}}\left[\psi_1(x_1)\,\psi_2(x_2) - \psi_1(x_2)\,\psi_2(x_1)\right]$$

2.6.3 Canonical Partition Functions

For N distinguishable particles with single-particle energy levels ϵ_i, and considering q to be the partition function for a single particle, we simply have:

$$Q = q^n = \left(\sum_i e^{-\beta\epsilon_i}\right)^N. \tag{2.110}$$

For indistinguishable particles it is better to work in terms of the occupation numbers $\{n_\nu\} = \{n_0, n_1, n_2, ...\}$. For *bosons* we consider the partition function to be:

$$Q = \sum_{\{n_i\}} \delta\left(\sum_i n_i - N\right) e^{-\beta\sum_i n_i\epsilon_i}. \tag{2.111}$$

For the case of *fermions* it reads:

$$Q = \sum_{\{n_i\}} \delta\left(\sum_i n_i - N\right)\left(\prod_i (\delta_{n_i,0} + \delta_{n_i,1})\right) e^{-\beta\sum_i n_i\epsilon_i}. \tag{2.112}$$

However, it is far from trivial to work analytically in the constant-N ensembles.

2.6.4 Grand-Canonical Partition Functions - Bose-Einstein, Fermi Statistics and Boltzmann Statistics

• We now consider the grand-canonical formalism to find explicit partition functions. We first consider the case of a general grand-canonical formalism by setting I to be some index set whose elements label single particle energy states and thus giving the partition function in a similar way as (2.66):

$$\begin{aligned}\Xi = \sum_{\{n_i\}} e^{-\beta\left[\sum_{i\in I} n_i\epsilon_i - \mu\sum_{i\in I} n_i\right]} = \sum_{\{n_i\}}\prod_{i\in I} e^{-\beta n_i(\epsilon_i-\mu)} &= \sum_{\{n_i\}} e^{-\beta n_1(\epsilon_1-\mu)} e^{-\beta n_2(\epsilon_2-\mu)}... \\ &= \left(\sum_{n_1=0}^{\infty} e^{-\beta n_1(\epsilon_1-\mu)}\right)\left(\sum_{n_2=0}^{\infty} e^{-\beta n_2(\epsilon_2-\mu)}\right)... \\ &= \prod_{i\in I}\sum_{n_i=0}^{\infty} e^{-\beta n_i(\epsilon_i-\mu)}.\end{aligned}$$

Thus we have a geometric series which yields:

$$\Xi = \prod_i \frac{1}{1-e^{-\beta(\epsilon_i-\mu)}} \Longrightarrow \ln(\Xi) = -\sum_i \left(1 - e^{-\beta(\epsilon_i-\mu)}\right). \tag{2.113}$$

Proposition 2.6.5. *The mean occupation numbers is given by:*

$$\langle n_j\rangle = \frac{1}{e^{\beta(\epsilon_j-\mu)}-1}. \tag{2.114}$$

Proof. This is a straightforward calculation:

$$\langle n_j \rangle = -\frac{\partial \ln(\Xi)}{\partial(\beta\epsilon_j)} = \frac{e^{-\beta(\epsilon_j-\mu)}}{1-e^{-\beta(\epsilon_j-\mu)}} = \frac{1}{e^{\beta(\epsilon_j-\mu)}-1}. \tag{2.115}$$

□

Now, the total particle number can be enforced by computing $\langle N \rangle$ and self-consistently adjusting μ.

• In the case of *fermions*, we can also work in the grand-canonical formalism, where $n_i \in \{0,1\}$. We then find:

$$\begin{aligned}
\Xi = \sum_{\{n_i\}} e^{-\beta\left[\sum_{i\in I} n_i\epsilon_i - \mu\sum_{i\in I} n_i\right]} = \sum_{\{n_i\}}\prod_{i\in I} e^{-\beta n_i(\epsilon_i-\mu)} &= \sum_{\{n_i\}} e^{-\beta n_1(\epsilon_1-\mu)} e^{-\beta n_2(\epsilon_2-\mu)} \ldots \\
&= \left(\sum_{n_1=0}^{1} e^{-\beta n_1(\epsilon_1-\mu)}\right)\left(\sum_{n_2=0}^{1} e^{-\beta n_2(\epsilon_2-\mu)}\right)\ldots \\
&= \prod_{i\in I}\sum_{n_i=0}^{1} e^{-\beta n_i(\epsilon_i-\mu)} \\
&= \prod_{i\in I}\left(1+e^{-\beta(\epsilon_i-\mu)}\right).
\end{aligned}$$

Thus we again have $\ln(\Xi) = \sum_i \ln\left(1+e^{-\beta(\epsilon_i-\mu)}\right)$.

Proposition 2.6.6. *The mean occupation numbers is given by:*

$$\langle n_j \rangle = \frac{1}{e^{\beta(\epsilon_j-\mu)}+1}. \tag{2.116}$$

Proof. Again this is a straightforward calculation:

$$\langle n_j \rangle = -\frac{\partial \ln(\Xi)}{\partial(\beta\epsilon_j)} = \frac{e^{-\beta(\epsilon_j-\mu)}}{1+e^{-\beta(\epsilon_j-\mu)}} = \frac{1}{e^{\beta(\epsilon_j-\mu)}+1}. \tag{2.117}$$

□

Also in this case, in the thermodynamic limit, constant-$\langle N \rangle$ may be set by adjusting μ.

• If we now consider the case of distinguishable particles for the grand-canonical ensemble one obtains:

$$\Xi = \sum_{N=0}^{\infty} q^N e^{\beta\mu N} = \frac{1}{1-qe^{\beta\mu}} \Longrightarrow \ln(\Xi) = -\ln\left(1-qe^{\beta\mu}\right). \tag{2.118}$$

Proposition 2.6.7. *The mean occupation numbers is given by:*

$$\langle n_j \rangle = \frac{e^{-\beta\epsilon_j}}{e^{-\beta\mu}-q}. \tag{2.119}$$

Proof. Again this is a straightforward calculation:

$$\langle n_j \rangle = -\frac{\partial \ln(\Xi)}{\partial(\beta\epsilon_j)} = -\frac{\partial \ln(\Xi)}{\partial q}\frac{\partial q}{\partial(\beta\epsilon_j)} = -\frac{e^{\beta\mu}}{1-qe^{\beta\mu}}\left(-e^{-\beta\epsilon_j}\right) = \frac{e^{-\beta(\epsilon_j-\mu)}}{1-qe^{\beta\mu}} = \frac{e^{-\beta\epsilon_j}}{e^{-\beta\mu}-q}. \tag{2.120}$$

□

Let us finally see how we can enforce the desired value for $\langle N \rangle$. In the high-temperature limit, for a system with an infinite number of energy levels, the number of levels gets much larger than $\langle N \rangle$. Then:

$$\langle n_j \rangle = \frac{1}{e^{\beta(\epsilon_j-\mu)} \pm 1} \ll 1 \Longrightarrow e^{\beta(\epsilon_j-\mu)} \gg 1. \tag{2.121}$$

Hence one can write $\langle n_j \rangle \approx e^{-\beta(\epsilon_j-\mu)}$ and so:

$$\langle N \rangle = \sum_j \langle n_j \rangle = e^{\beta\mu}\sum_j e^{-\beta\epsilon_j} \Longrightarrow \beta\mu = \ln\left(\langle N \rangle\right) - \ln\left(\sum_j e^{-\beta\epsilon_j}\right). \tag{2.122}$$

Let us now apply a few thermodynamic relations to find $\ln(Q)$:

$$\begin{aligned}
\ln(Q) = -\beta A = -\beta(G-\rho V) = -\beta\mu\langle N \rangle + \ln(\Xi) &= -\beta\mu\langle N \rangle \pm \sum_i \ln\left(1 \pm e^{-\beta(\epsilon_i-\mu)}\right) \\
&\approx -\beta\mu\langle N \rangle + e^{\beta\mu}\sum_i e^{-\beta\epsilon_i} \\
&= -\langle N \rangle \ln(\langle N \rangle) + \langle N \rangle \ln\left(\sum_j e^{-\beta\epsilon_j}\right) + \langle N \rangle \\
&\approx -N! + \langle N \rangle \ln(q).
\end{aligned}$$

Thus, the high-temperature partition function is equal to the Boltzmann case with a correction to account for the indistinguishability of the particles:

$$Q = \frac{q^N}{N!}. \tag{2.123}$$

Chapter 3

Models in Statistical Mechanics and Applications in Materials Science

3.1 Vibrations and Electronic Excitations in Solids

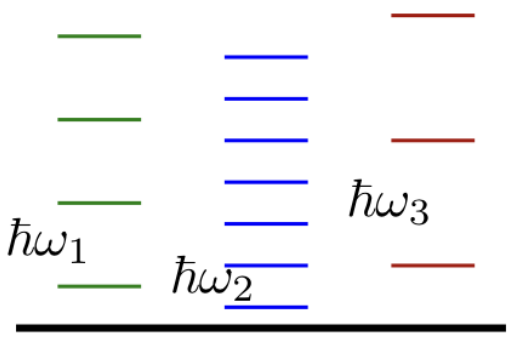
Consider a solid with its atoms in their equilibrium positions $\mathbf{x}_i^{(0)}$. The energy for small displacements can be written as a Taylor expansion around the equilibrium positions:

$$V(\mathbf{x}) = V\left(\{\mathbf{x}_i^{(0)}\}\right) + \underbrace{\sum_i \left(\mathbf{x}_i - \mathbf{x}_i^{(0)}\right) \frac{\partial V}{\partial \mathbf{x}_i}}_{=0} + \frac{1}{2} \sum_{i,j=1}^{N} \left(\mathbf{x}_i - \mathbf{x}_i^{(0)}\right) \frac{\partial^2 V}{\partial \mathbf{x}_i \partial \mathbf{x}_j} \left(\mathbf{x}_j - \mathbf{x}_j^{(0)}\right), \tag{3.1}$$

where we have used the fact that at the equilibrium position $\frac{\partial V}{\partial \mathbf{x}_i}|_{\mathbf{x}_i = \mathbf{x}_i^{(0)}} = 0$. This is a quadratic form in $3N$ dimensions with the different coordinates coupled by $H_{ij} = \frac{\partial^2 V}{\partial \mathbf{x}_i \partial \mathbf{x}_j}$. We may diagonalize the Hessian matrix $\mathbf{H}$ and rewrite the potential in the basis of its eigenvectors, with the normal mode coordinates u_k. This is an eigendecomposition $\mathbf{H}\underline{e}_k = \omega_k^2 \underline{e}_k$, and then set $u_k = \left(\mathbf{x} - \mathbf{x}^{(0)}\right) \cdot \underline{e}_k$. The u_k corresponds to a displacement along the phonon. Thus:

$$V(\mathbf{u}) = V_0 + \frac{1}{2} \sum_k m \omega_k^2 u_k^2. \tag{3.2}$$

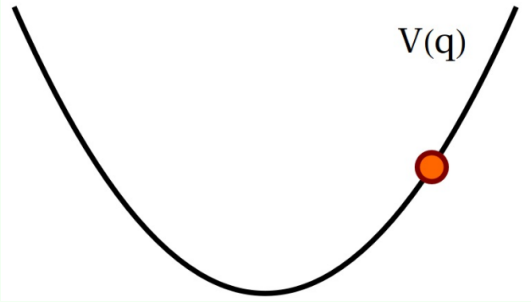
Each normal mode corresponds to a separate system where the thermodynamics of the solid can be seen as a combination of independent harmonic oscillators.



Example 3.1.1 (A classical oscillator)**.** *Let us compute the canonical partition function of a harmonic oscillator of frequency ω:*

$$Q = \int e^{-\beta\left(\frac{1}{2}m\omega^2 q^2 + \frac{p^2}{2m}\right)} dqdp = \frac{2\pi}{\sqrt{m\omega^2\beta}\sqrt{\beta/m}} = \frac{2\pi}{\beta\omega}. \tag{3.3}$$

From there it is elementary to find the usual physical quantities. The free energy A is given by $A = -k_B T \ln(Q) = cst. + k_B T \ln(\beta\omega)$, the mean energy is given by $\langle E\rangle = -\frac{\partial \ln(Q)}{\partial\beta} = \frac{1}{\beta}$ and finally, the heat capacity is given by (Dulong-Petit Law) $C_v = \frac{\partial\langle E\rangle}{\partial T} = k_B$.



Example 3.1.2 (A quantum oscillator)**.** *Let us compute the canonical partition function of a quantum oscillator with energy levels $\hbar\omega\left(\frac{1}{2}+k\right)$:*

$$Q = \sum_{k=0}^{\infty} e^{-\beta\hbar\omega\left(\frac{1}{2}+k\right)} = e^{-\frac{\beta\hbar\omega}{2}} \sum_{k=0}^{\infty} e^{-\beta\hbar\omega k} = \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1-e^{-\beta\hbar\omega}} = \frac{1}{2}\left[\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right]^{-1}. \tag{3.4}$$

From there it is elementary to find the usual physical quantities. The free energy A is given by:

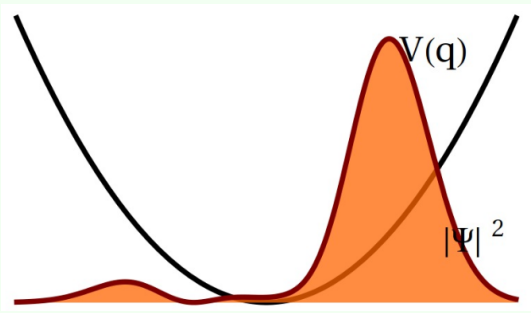
$$A = -k_B T \ln(Q) = \frac{\hbar\omega}{2} + k_B T \ln\left(1 - e^{-\beta\hbar\omega}\right). \tag{3.5}$$

The mean energy is given by:

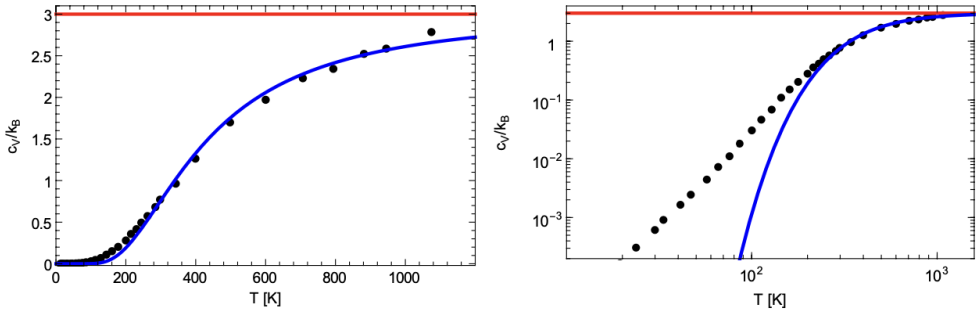
$$\langle E\rangle = -\frac{\partial \ln(Q)}{\partial\beta} = \frac{\hbar\omega}{2} + \hbar\omega\frac{e^{-\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}} = \frac{\hbar\omega}{2}\coth\left(\frac{\beta\hbar\omega}{2}\right). \tag{3.6}$$

Finally, the heat capacity is given by:

$$C_v = \frac{\partial\langle E\rangle}{\partial T} = k_B\left[\frac{2}{\beta\hbar\omega}\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right]^{-2}. \tag{3.7}$$



An alternative view on the problem is to consider each excitation as one quasiparticle, occupying the energy level $\hbar\omega$ and subtracting the zero-point energy $\frac{\hbar\omega}{2}$. In this sense, each vibrational modes contributes to the overall partition function a term of $\ln(Q) = \ln\left(1 - e^{-\beta\hbar\omega}\right)$ that is equivalent to the partition function for bosons with the chemical potential set to zero - these yield *phonons*.
For instance, the heat capacity of diamond, for example, has a classical value of $C_v = 3k_B$ (Dulong-Petit). If we look at $3N$ quantum oscillators at frequency ω_E, which corresponds to an *Einstein solid*, everything looks good. However we have a wrong behavior at low temperature.



3.1.1 The Debye Solid

The Debye model provides a more refined approach to understanding heat capacity in solids by treating thermal vibrations of the atomic lattice as phonons confined in a box. This contrasts with the Einstein model, which assumes the solid is made up of non-interacting quantum harmonic oscillators (QHO), each vibrating independently. A key success of the Debye model is its ability to accurately describe the behavior of solids at low temperatures. Specifically, it predicts that the heat capacity C_v is proportional to T^3 (the Debye T^3 law), which matches experimental observations. At high temperatures, both the Debye and Einstein models approach the Dulong–Petit law, where the heat capacity levels off to a constant value per atom. However, the Debye model relies on certain simplifying assumptions, such as treating the vibrational spectrum as continuous, which causes its predictions to be less accurate at intermediate temperatures.
One should consider the contribution from different phonon frequencies. For a Debye solid with a phonon density of states $g(\omega)$, the specific heat is given by:

$$C_v(T) = \int_0^{\omega_D} g(\omega) C_V^{QHO}(\omega, T) d\omega. \tag{3.8}$$

At low-frequency elastic vibrations, the dispersion is linear $\omega(k) \propto k$ and in 3-dimensions the density of states is $g(k) \propto k^2 \Longrightarrow g(\omega) \propto \omega^2$. We now perform the change of variable $x = \frac{\hbar\omega}{k_B T} \Longrightarrow d\omega = \frac{k_B T}{\hbar} dx$. Thus:

$$C_v(T) \propto \int_0^{\omega_D} \omega^2 \left[\frac{k_B T}{\hbar\omega} \sinh\left(\frac{\hbar\omega}{2k_B T}\right)\right]^{-2} d\omega = \int_0^{\frac{\hbar\omega_D}{k_B T}} \left(\frac{k_B T}{\hbar}\right)^3 x^2 \left[\frac{1}{x} \sinh\left(\frac{x}{2}\right)\right]^{-2} dx. \tag{3.9}$$

Therefore, as $T \to 0$ we have that $C_v(T) \propto T^3 \int_0^{\infty} x^4 \sinh^{-2}\left(\frac{x}{2}\right) dx$, which is exactly the result of the Debye T^3 law.

3.1.2 Thermal Properties of Graphene

Graphene has very unusual thermal properties that can to be understood based on its phonon dispersion curves. It contains two modes with $\omega(k) \propto k$ and one branch corresponding to $\omega(k) \propto k^2$. The

branch in question is the out-of-plane ZA branch which describes the perpendicular displacement of atoms occurs to the direction of wave propagation. In 2D materials, this mode is associated with out-of-plane vibrations.

Let us first ignore the ZA modes, and we consider a 2-dimensional linear dispersion with density of states $g(k) \propto k \implies g(\omega) \propto \omega$. The heat capacity is given by:

$$C_v^{2D}(T) \propto \int_0^{\omega_D} \omega \left[\frac{k_B T}{\hbar\omega} \sinh\left(\frac{\hbar\omega}{2k_B T}\right)\right]^{-2} d\omega \propto T^2 \int_0^{\frac{\hbar\omega_D}{k_B T}} x^3 \sinh^{-2}\left(\frac{x}{2}\right) dx. \tag{3.10}$$

Thus, in the absence of ZA modes, the heat capacity is proportional to T^2.

Let us now include the ZA modes, and we consider a 2-dimensional quadratic dispersion $\omega(k) \sim k^2 \implies d\omega = kdk$ with density of states $g(k)dk \propto kdk \implies g(\omega) \propto 1$. The heat capacity is given by:

$$C_v^{2D,ZA}(T) \propto \int_0^{\omega_D} \left[\frac{k_B T}{\hbar\omega} \sinh\left(\frac{\hbar\omega}{2k_B T}\right)\right]^{-2} d\omega \propto T \int_0^{\frac{\hbar\omega_D}{k_B T}} x^2 \sinh^{-2}\left(\frac{x}{2}\right) dx. \tag{3.11}$$

Thus, in the presence of ZA modes, the heat capacity is proportional to T.

3.1.3 The Free Electron Gas

Free electrons are characterized by pure kinetic energy (no potential energy), with eigenstates labelled by a wavevector $\mathbf{k}$ and energy $\frac{\hbar^2\mathbf{k}^2}{2m_e}$, where m_e is the electron mass. If we consider a cubic volume $V = L^3$ with periodic boundary conditions, states are quantized and characterized by $\mathbf{k} = \frac{2\pi}{L}(n_x, n_y, n_z)$. For large L and some quantity g, sums over energy levels can be approximated as integrals:

$$\sum_{\mathbf{n},\sigma} g = 2 \cdot \frac{V}{8\pi^3} \int g d\mathbf{k} = \frac{V}{4\pi^3} \int g \cdot 4\pi k^2 dk = \frac{V}{\pi^2} \int g \cdot k^2 dk. \tag{3.12}$$

The total number of electrons is set by (3.12):

$$\langle N \rangle = \frac{V}{\pi^2} \int \frac{1}{1 + \exp\left[\beta\left(\frac{\hbar^2 k^2}{2m} - \epsilon_F\right)\right]} k^2 dk. \tag{3.13}$$

As $T \to 0$, the Fermi function behaves as a step function going from 1 to 0 for $\hbar^2 k_F^2 = 2m_{\epsilon_F}$. Hence:

$$\frac{\langle N \rangle}{V} = \rho = \frac{1}{3\pi^2} k_F^3 \implies k_F = (3\pi^2\rho)^{\frac{1}{3}}. \tag{3.14}$$

Again by (3.12), the energy density $\frac{\langle E \rangle}{V}$ may be given as a function ρ:

$$\langle E \rangle = \frac{V}{\pi^2} \int_0^{k_F} \frac{\hbar^2 k^2}{2m} k^2 dk = \frac{V}{\pi^2} \frac{\hbar^2}{2m} \frac{k_F^5}{5} \implies \frac{\langle E \rangle}{V} = \frac{\hbar^2 k_F^5}{10m\pi^2} = \frac{3\hbar^2 (9\pi^4)^{\frac{1}{3}}}{10m} \rho^{\frac{5}{3}}. \tag{3.15}$$

The energy of the free electron gas is a function of the density ρ. This expression constitutes the basis for the Thomas-Fermi version of *density functional theory*.

We finally make a comment on how to go beyond the fact of considering the Fermi function as a step behaving function. A general way to express integrals of the Fermi function is using a *Sommerfeld expansion*. The general expansion, for $\beta^{-1} = k_B T$, is given by:

$$\int_{-\infty}^{\infty} \frac{H(\epsilon)}{1+e^{\beta(\epsilon-\epsilon_F)}} d\epsilon = \int_{-\infty}^{\epsilon_F} H(\epsilon) d\epsilon + \frac{\pi^2}{6} H'(\epsilon_F) (k_B T)^2 + \mathcal{O}\left[\left(\frac{k_B T}{\epsilon_F}\right)^4\right] \equiv cst. + \frac{\pi^2}{6} H'(\epsilon_F) (k_B T)^2. \tag{3.16}$$

Let us now consider a general density of states $g(\epsilon)$ and a finite temperature T:

$$\frac{\langle E \rangle}{V} = \int \frac{\epsilon g(\epsilon)}{1+e^{\beta(\epsilon-\epsilon_F)}} d\epsilon. \tag{3.17}$$

Then applying a Sommerfeld expansion (3.16) to $H(\epsilon) = \epsilon g(\epsilon)$ yields:

$$\frac{\langle E \rangle}{V} = cst. + \frac{\pi^2}{6} \left(\epsilon_F g'(\epsilon_F) + g(\epsilon_F)\right) (k_B T)^2. \tag{3.18}$$

And thus it is clear that the heat capacity in a metal at low temperature is proportional to T since:

$$C_v(T) := \frac{\partial \langle E \rangle}{\partial T} = \frac{\partial}{\partial T} \left(cst. + \frac{\pi^2}{6} \left(\epsilon_F g'(\epsilon_F) + g(\epsilon_F)\right) V (k_B T)^2 \right) \propto T. \tag{3.19}$$

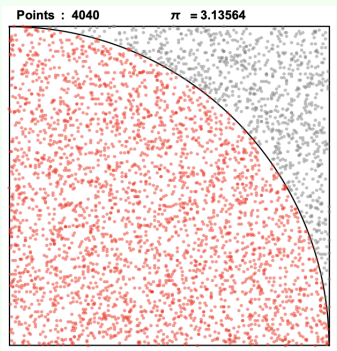
3.2 Simulation and Sampling (Not Examinable)

The canonical ensemble describes a system in contact with a thermal reservoir at temperature $T = \frac{1}{k_B \beta}$. Averages of observables can be computed as integrals over the canonical Boltzmann distribution:

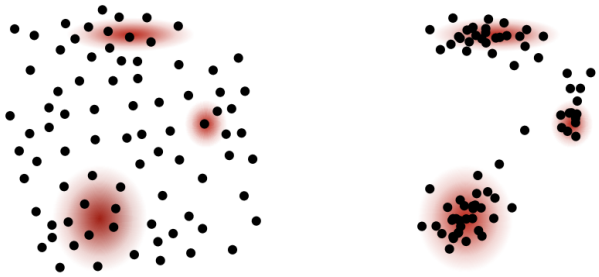
$$\langle A \rangle = \int \mathbb{P}(\mathbf{p}, \mathbf{q}) A(\mathbf{p}, \mathbf{q}) d\mathbf{q} d\mathbf{p} \quad \text{or} \quad \langle A \rangle = \sum_{\nu} A_\nu \mathbb{P}_\nu.$$

The goal of a simulation is to perform these averages/integrals by sampling the states that have a non-negligible probability. Then since statistical averaging is basically integration, it shouldn't be difficult to implement numerically. Performing the integral by quadrature is impractical since the number of grid points grows exponentially with dimensionality. For instance let us think about computing averages for a spin lattice with N sites. Enumeration of all possible spin states also grows exponentially with 2^N possible configurations must be traversed to cover all possibilities. However, often only a small fraction of microstates have $\mathbb{P}_\nu \gg 0$ and also we don't need to sample all states but only a representative fraction of them.

Example 3.2.1 (Monte Carlo Integration)**.** *One can compute the area of a circle relative to an enclosing square by tossing random values on a square and counting how many have $x^2 + y^2 < 1$ (this fraction converges to $\frac{\pi}{4}$).*



As we have seen in the previous example, to compute integrals without a grid, we may use Monte Carlo integration to generate uniformly distributed random values and compute the value of the function in the region. However, we can also compute $\int f(x)dx$ as $area \times \frac{1}{N}\sum_i f(x_i)$. One could think of using the same trick for randomly-distributed points $\mathbf{q}_i$ and compute $\langle A\rangle \approx \sum_i \frac{\mathbb{P}(\mathbf{q}_i)A(\mathbf{q}_i)}{\sum_i \mathbb{P}(\mathbf{q}_i)}$. However, $\mathbb{P}(\mathbf{q})$ is often very irregular and most points do not contribute to the integral. Ideally, one would want to generate $\mathbf{q}_i$'s distributed as $\mathbb{P}(\mathbf{q})$ and then compute $\langle A\rangle \approx \frac{1}{M}\sum_i A(\mathbf{q}_i)$. This method is called *importance sampling*.



The underlying assumption behind all this is *ergodicity*:

$$\lim_{M\to\infty}\frac{1}{M}\sum_i A(\mathbf{q}_i) = \int \mathbb{P}(\mathbf{q})A(\mathbf{q})d\mathbf{q}. \tag{3.20}$$

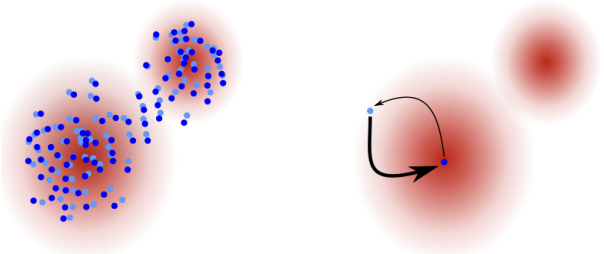
This condition is generally hard to prove in practice. However, the key tool to assess the ergodicity of a trajectory for computing $\langle A\rangle$ is the autocorrelation function [see Remark (1.2.11)] $c_{AA}(t) = \langle A(0), A(t)\rangle$.

Let us now describe how to use Monte Carlo for importance sampling. To generate a sequence of points distributed as $\mathbb{P}(\mathbf{q})$ we use Monte Carlo moves $\mathbf{q}\to\mathbf{q}'$ that generates a new point given a previous point consistent with $\mathbb{P}$. The move typically contains some randomness that is included in the probability $p(\mathbf{q}\to\mathbf{q}')$. The **necessary** and sufficient condition for invariance of $\mathbb{P}$ is that:

$$\int \mathbb{P}(\mathbf{q})p(\mathbf{q}\to\mathbf{q}')d\mathbf{q} = \mathbb{P}(\mathbf{q}'). \tag{3.21}$$

However, a **sufficient** and easier condition is given by **detailed balance**, which guarantees that the target distribution $\mathbb{P}(\mathbf{q})$ is conserved:

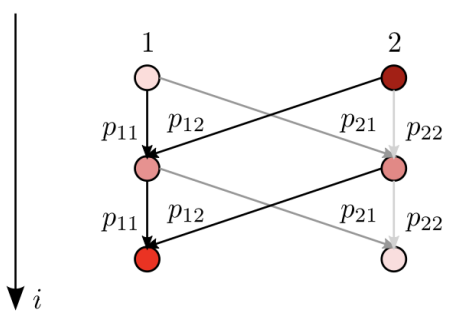
$$\mathbb{P}(\mathbf{q})p(\mathbf{q}\to\mathbf{q}') = \mathbb{P}(\mathbf{q}')p(\mathbf{q}'\to\mathbf{q}). \tag{3.22}$$



Example 3.2.2 (Starting from the wrong distribution)**.** *Consider a two state problem with equilibrium distributions* $\mathbb{P}_1 = 0.1$ *and* $\mathbb{P}_2 = 0.9$*. Take* $p(1 \to 2) = 0.45$*,* $p(1 \to 1) = 0.55$*,* $p(2 \to 1) = 0.05$ *and* $p(2 \to 2) = 0.95$*. Say we are in state 1 with probability* x_1 *and in 2 with probability* x_2*. Let* $p_{ij} = p(j \to i)$*. One Monte Carlo step is equivalent to a matrix-vector product* $\mathbf{x}' \leftarrow \mathbf{p}\mathbf{x}$*. The repetition of* N *applications of* $\mathbf{p}$ *is* $\mathbf{p}^N$ *that we can characterize in terms of the* N*-th power of its eigenvalues. The matrix for* $\mathbf{p}$ *is given by:*

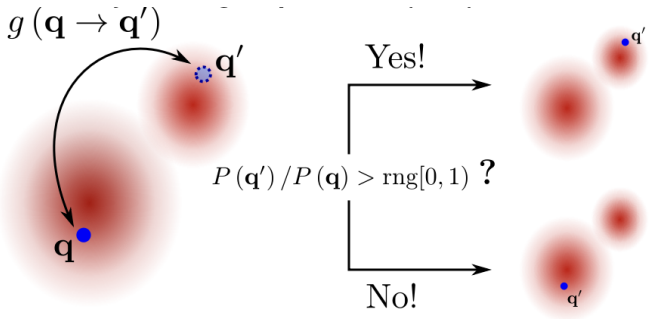
$$\mathbf{p} = \begin{pmatrix} 0.8 & 0.3 & 0.5 \\ 0.1 & 0.3 & 0.1 \\ 0.1 & 0.4 & 0.4 \end{pmatrix}. \tag{3.23}$$

One can build a transition matrix that violates detailed balance but still has a well-defined stationary distribution!



Theorem 3.2.3 (Metropolis moves - an algorithm)**.** *A simple way to generate moves consistent with detailed balance is to split the move in two steps: generation/acceptance. Thus set* $p(\mathbf{q} \to \mathbf{q}') = g(\mathbf{q} \to \mathbf{q}')a(\mathbf{q} \to \mathbf{q}')$ *and we follow the following steps:*

- *Generate a new configuration with a criterion that is symmetric:* $g(\mathbf{q} \to \mathbf{q}') = g(\mathbf{q}' \to \mathbf{q})$*.*
- *Accept a new configuration with probability* $a(\mathbf{q} \to \mathbf{q}') = \min\left[1, \frac{\mathbb{P}(\mathbf{q}')}{\mathbb{P}(\mathbf{q})}\right] = \min\left[1, e^{-\beta\left(V(\mathbf{q}')-V(\mathbf{q})\right)}\right]$*.*
- *In case of a reject, add* $\mathbf{q}$ *again to the list of samples.*

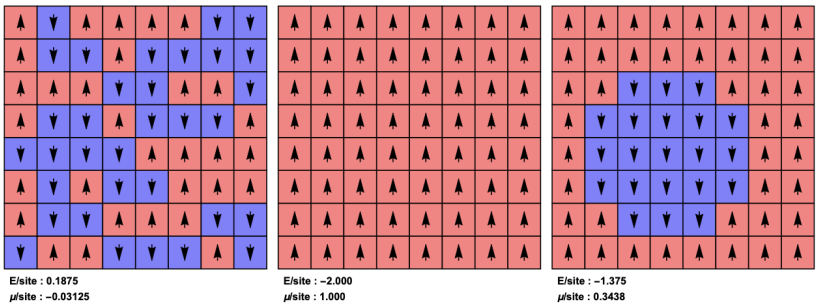


Metropolis et al., J. Chem. Phys 1953

One could also use sampling by molecular dynamics, in particular using Hamilton's equations to generate a trajectory. This will not obey detailed balance anymore, but rather a generalization of it. Further details are left to the interest of the reader.

3.3 Ising Model

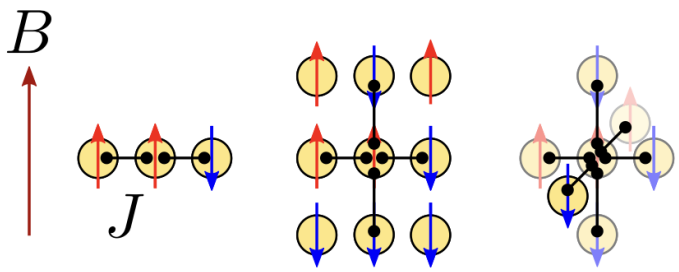
Recall that for a paramagnetic material there is no magnetization in the absence of an external field. We now present the Ising Model, which is a model of ferromagnetism, which corresponds to the situation where there is a residual magnetization at zero field. The microscopic model is similar to that of a paramagnet. Each atom carries a magnetic moment that can orient itself relative to a magnetic field. A crucial point is that spins interact with each other, hence in general $Q \neq q^N$.



The Ising Model is the simplest model of spin interactions, where each site is associated with a number $s_j = \pm 1$ and the first-neighbor interactions are given by the Heisenberg Hamiltonian

$$H = -\mu_B B \sum_i s_i - \frac{1}{2} J \sum_i \sum_{j \in \mathcal{N}_i} s_i s_j. \tag{3.24}$$

Here, $\mathcal{N}_i$ is the set of neighboring spins of the spin i. The behavior of this model depends dramatically on the dimension of the lattice that we are considering.



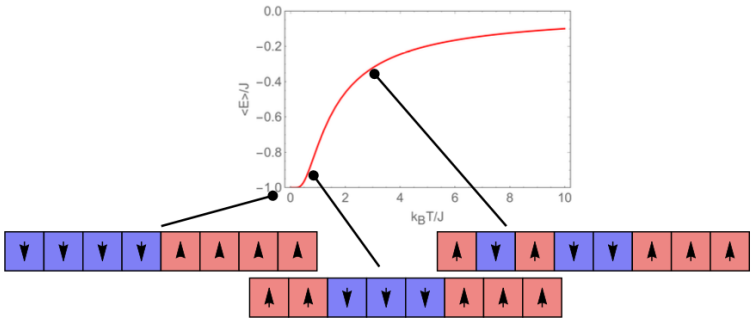
3.3.1 Ising Model in 1-Dimension

Consider the case without a magnetic field. We can make a change of variable $s_i s_{i+1} \longrightarrow s'_i$ and we obtain:

$$Q = \sum_{\{\mathbf{s}\}} \prod_i e^{\beta J s_i s_{i+1}} = \sum_{\{\mathbf{s}'\}} \prod_i e^{\beta J s'_i} = \prod_i \left(e^{\beta J} + e^{-\beta J}\right) = \left[2\cosh(\beta J)\right]^N.$$

The average energy is given by:

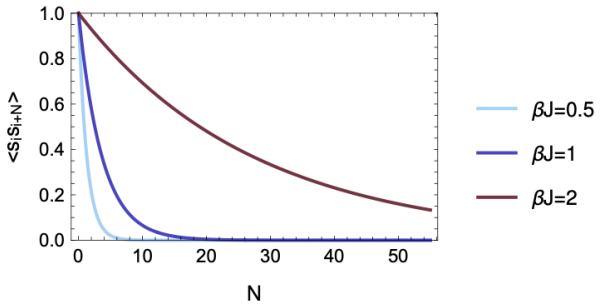
$$\langle E \rangle = -\frac{\partial \ln(Q)}{\partial \beta} = -NJ \frac{\left(e^{J\beta} - e^{-J\beta}\right)}{\left(e^{J\beta} + e^{-J\beta}\right)} = -NJ \tanh(\beta J). \tag{3.25}$$



Now recall that $\langle s_i s_{i+N} \rangle$ measures the probability that two spins separated by N sites have the same orientation. By making the substitution $s_i s_{i+1} \longrightarrow s'_i$ we have that:

$$\langle s_i s_{i+N} \rangle = \frac{\sum_{\{\mathbf{s}\}_N} s_1 s_N \prod_i e^{\beta J s_i s_{i+1}}}{\sum_{\{\mathbf{s}\}_N} \prod_i e^{\beta J s_i s_{i+1}}} = \frac{\sum_{\{\mathbf{s'}\}_N} \prod_i s'_i e^{\beta J s'_i}}{\sum_{\{\mathbf{s'}\}_N} \prod_i e^{\beta J s'_i}} = \left[\tanh(\beta J)\right]^N. \tag{3.26}$$

Thus we obtain an exponential decay of correlations with no phase transition, except at $T = 0$.



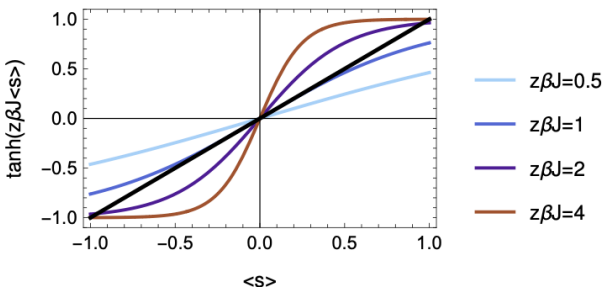
3.3.2 Ising Model in 2-Dimensions and Above - The Mean Field Model

Consider the substitution $s_i \longrightarrow \langle s \rangle + \delta s_i$ and neglect the quadratic term $\delta s_i \delta s_j$. The partition function can be factorized where the probability depends on the mean spin $\langle s \rangle$ and the number of neighbors z:

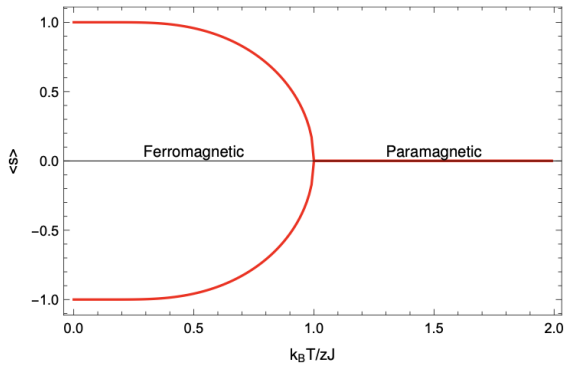
$$\begin{aligned} -H_{MF} &= \sum_i \mu_B B(\langle s \rangle + \delta s_i) + \frac{1}{2} J \sum_i \sum_{j \in \mathcal{N}_i} (\langle s \rangle + \delta s_i)(\langle s \rangle + \delta s_j) \\ &= \sum_i \mu_B B(\langle s \rangle + \delta s_i) - \frac{1}{2} z J \langle s \rangle^2 + z J \langle s \rangle (\langle s \rangle + \delta s_i) \\ &= -\frac{N}{2} z J \langle s \rangle^2 + \sum_i \mu_B B s_i + z J \langle s \rangle s_i, \qquad \text{where } \mathbb{P}(s_i) \propto e^{\beta[\mu_B B + z J \langle s \rangle] s_i}. \end{aligned}$$

Now, the self-consistent equation for $\langle s \rangle$ is given by:

$$\langle s \rangle = \sum_{\pm 1} s_i \mathbb{P}(s_i) = \tanh\left(\beta \mu_B B + z J \beta \langle s \rangle\right). \tag{3.27}$$



Observe that for $B = 0$, $\langle s \rangle = \tanh\left(\beta\mu_B B + zJ\beta\langle s \rangle\right)$ has two solutions for $\frac{k_B T}{zJ} < 1$. For $\frac{k_B T}{zJ} > 1$ has only one solution for $\langle s \rangle = 0$. Finally note that the system has a phase transition at $T_c = \frac{zJ}{k_B}$.



3.3.3 Magnetization

Let us now look at the *magnetization*. We start by considering the partition function:

$$Q = q^N = \left[2e^{-\frac{\beta z J \langle s \rangle^2}{2}} \cosh\left(\beta\mu_B B + \beta z J \langle s \rangle\right)\right]^N. \tag{3.28}$$

Recall that the magnetization is originally defined as $M = \sum_i s_i$. The mean magnetization $\langle M \rangle$ is:

$$\langle M \rangle = \mu_B N \langle s \rangle = \frac{1}{\beta}\frac{\partial \ln(Q)}{\partial B} = N\mu_B \tanh\left(\beta\mu_B B + \beta z J \langle s \rangle\right). \tag{3.29}$$

At high temperature with $\langle s \rangle \approx 0$, a $\mathcal{O}(B)$ expansion yields a paramagnet. The magnetization per spin is given by:

$$\langle m \rangle = \frac{\langle M \rangle}{N} = \mu_B \langle s \rangle = \frac{\mu_B^2 B}{k_B T}. \tag{3.30}$$

Close to the critical point and for $B = 0$, we may take a series expansion for $\tanh(x) \approx x - \frac{1}{3}x^3$. By defining the reduced temperature by $t = \frac{T - T_c}{T_c}$ we have:

$$\langle s \rangle^2 = \left[\frac{T_c}{T}\langle s \rangle - \frac{1}{3}\left(\frac{T_c}{T}\right)^3 \langle s \rangle^3\right]^2 = 3\left(\frac{T}{T_c}\right)^3 \left(\frac{T_c}{T} - 1\right) = 3(1+t)^2(-t) \approx -3t.$$

Thus, the order parameter of $\langle M \rangle$ grows as $t^{\frac{1}{2}}$ close to T_c. In fact, this is an example of a *critical exponent*, describing the behavior close to a phase transition.

3.3.4 Susceptibility

The response to a magnetic field is given by the *susceptibility* $\chi = \frac{\partial \langle m \rangle}{\partial B}$. Taking first order expansion in B of $\langle m \rangle$ yields:

$$\langle m \rangle = \beta\mu_B^2 B + \frac{T_c}{T}\langle m \rangle - \frac{1}{3\mu_B^2}\left(\frac{T_c}{T}\right)^3 \langle m \rangle^3. \tag{3.31}$$

Thus we have:

$$\chi = \beta\mu_B^2 + \frac{T_c}{T}\chi - \left(\frac{T_c}{T}\right)^3 \chi \langle s \rangle^2. \tag{3.32}$$

Now, for $T \gtrsim T_c$ (T is greater than or roughly equal to T_c) with $\langle m \rangle \approx 0$:

$$\chi \approx \frac{\beta \mu_B^2}{1 - \frac{T_c}{T}} \approx \frac{\beta \mu_B^2}{t}. \tag{3.33}$$

And for $T \lesssim T_c$ we retain the $\mathcal{O}\left(\langle s \rangle^2\right)$ terms and find:

$$\chi \approx \frac{\beta \mu_B^2}{1 - \frac{T_c}{T} + 3\left(\frac{T}{T_c}\right)^3 \left(\frac{T_c}{T} - 1\right)\left(\frac{T_c}{T}\right)^3} = \frac{\beta \mu_B^2}{1 - \frac{T_c}{T} + 3\left(\frac{T_c}{T} - 1\right)} \approx -\frac{\beta \mu_B^2}{2t}. \tag{3.34}$$

In these instances, the critical exponent (i.e. exponent above t) for χ is 1.

3.3.5 Final Remarks and Mappings

• The mean-field model has some limitations. It predicts a finite temperature phase transition for the 1-dimensional Ising Model, however we have an overestimation of the transition temperature in higher dimensions. We may obtain several better (but rather more complex) approximations, or one may use explicit simulations of the Ising Model for the analysis.

• To analyze models (e.g. a lattice gas) we may use a mapping on the Ising Model. For instance, the Hamiltonian describes small cells that have either 0 or 1 molecules. The chemical potential is μ_B and the bond energy is ϵ and thus:

$$H = -\mu_B \sum_i n_i - \frac{\epsilon}{2} \sum_{i,j \in \mathcal{N}_i} n_i n_j. \tag{3.35}$$

This can be mapped on an Ising Model with the substitution $n_i = \frac{s_i + 1}{2}$, setting $\mu_B B = \frac{\mu_B}{2} + z\frac{\epsilon}{4}$ and $J = \frac{\epsilon}{4}$:

$$H = C - \mu_B B \sum_i s_i - \frac{J}{2} \sum_{i,j \in \mathcal{N}_i} s_i s_j. \tag{3.36}$$

3.4 Statistical Mechanics of Liquids

A theory of the liquid state is an important topic for Materials Science (e.g. medium for synthesis, processing in the molten phase, models for glasses,...). In general, liquids pose a challenge to modeling and for a theoretical description. Atoms are disordered hence there is no way to invoke a harmonic approximation and atoms are also strongly interacting and thus we cannot invoke independent particles approximations.

3.4.1 Classical Methods, Reduced Distribution Functions and the Pair Correlation Function

• Recall that atoms are treated as classical particles following the Hamiltonian

$$H = K(\mathbf{p}) + U(\mathbf{r}) = \sum_i \frac{\mathbf{p}_i^2}{2m_i} + U(\{\mathbf{r}_i\}) \tag{3.37}$$

with positions $\mathbf{r} = \{\mathbf{r}_i\}$ and momenta $\mathbf{p} = \{\mathbf{p}_i\}$. The potential is a rather complicated, non-linear function of all the coordinates. However, the partition function Q factors nicely in a free-particle part

(consider the classical limit of the quantum counterpart) and one that depends on U:

$$Q = Q_{\mathbf{p}} Q_{\mathbf{r}} = \frac{1}{N!}\left[\frac{h}{\sqrt{2\pi m k_B T}}\right]^{3N} \int e^{-\beta U(\{\mathbf{r}_i\})} d^N\{\mathbf{r}_i\}. \tag{3.38}$$

And thus the probability of observing a given configuration of the atoms is just

$$\mathbb{P}(\mathbf{r}_1, ..., \mathbf{r}_N) = \frac{1}{Q_{\mathbf{r}}} e^{-\beta U(\{\mathbf{r}_i\})}. \tag{3.39}$$

• Now, we can always marginalize $\mathbb{P}$ with respect to some of the atoms. Take for instance a single-particle marginal. Since all the atoms are statistically equivalent, it is more convenient to define a generic distribution function summed over all particle indices, called the **reduced distribution function**:

$$\mathbb{P}^{(1)}(\mathbf{r}_1) = \int \mathbb{P}(\mathbf{r}_1, ..., \mathbf{r}_N) d^{N-1}(\mathbf{r}_2, ..., \mathbf{r}_N). \tag{3.40}$$

We may also set $\rho^{(m)}$ to be the probability that any set of m particles occupies the positions $\mathbf{r}$. Then, for a single particle we obtain

$$\rho^{(1)}(\mathbf{r}) = \sum_i \mathbb{P}^{(1)}(\mathbf{r}_i) = N\mathbb{P}^{(1)}(\mathbf{r}_1 = \mathbf{r}), \tag{3.41}$$

and if we are in the case of having no external potential, it yields $\rho^{(1)}(\mathbf{r}) = N \cdot \frac{1}{V} = \rho$ [Density of states!]. Let us now look at the two-particle case. The reduced distribution function is given by

$$\mathbb{P}^{(2)}(\mathbf{r}_1, \mathbf{r}_2) = \int \mathbb{P}(\mathbf{r}_1, ..., \mathbf{r}_N) d^{N-2}(\mathbf{r}_3, ..., \mathbf{r}_N), \tag{3.42}$$

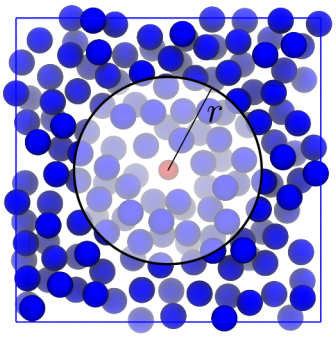
where the probability that any set of 2-particles occupies the positions $\mathbf{r}, \mathbf{r}'$ is

$$\rho^{(2)}(\mathbf{r}, \mathbf{r}') = \sum_{i \neq j} \mathbb{P}^{(2)}(\mathbf{r}_i, \mathbf{r}_j) = N(N-1)\mathbb{P}^{(2)}(\mathbf{r}_1 = \mathbf{r}, \mathbf{r}_2 = \mathbf{r}'). \tag{3.43}$$

For instance, if we consider an ideal gas we have $\rho^{(2)} = \frac{N(N-1)}{V^2} = (1 - N^{-1})\rho^2$.

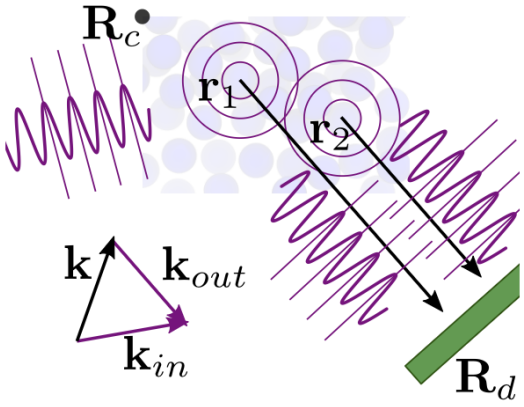
• In an isotropic medium (i.e. a material where properties like permittivity, permeability and refractive index are the same in all the directions), $\rho^{(2)}(\mathbf{r}, \mathbf{r}')$ depends only on $r = |\mathbf{r} - \mathbf{r}'|$. We can then define a normalized version of $\rho^{(2)}$ that indicates the probability of finding a particle at distance r from the origin, conditional on having one particle at 0. This is called the **pair correlation function** $g(r)$ and it is defined by:

$$g(r) := \frac{1}{\rho^2} \rho^{(2)}(\mathbf{0}, \mathbf{r}). \tag{3.44}$$



3.4.2 An Application - Scattering from a Liquid

Consider the scattering/diffusion of X-rays (or similarly, neutrons) from the atoms. Scattering is nearly elastic hence



Scattering is nearly elastic hence $k_{in} = k_{out}$. Define the difference of momenta $\mathbf{k} = \mathbf{k}_{out} - \mathbf{k}_{in}$, where $\mathbf{k}_{out}$ and $\mathbf{k}_{in}$ are the out and in momenta respectively. The amplitude of the waves scattered by an atom at position $\mathbf{r}_i$ relative to the center of the sample $\mathbf{R}_c$ reads:

$$A_i \propto e^{i\mathbf{k}_{in}(\mathbf{R}_c+\mathbf{r}_i)} e^{i\mathbf{k}_{out}(\mathbf{R}_d-(\mathbf{R}_c+\mathbf{r}_i))} = e^{i(\mathbf{k}_{in}-\mathbf{k}_{out})\mathbf{R}_c} e^{i\mathbf{k}_{out}\mathbf{R}_d} e^{i(\mathbf{k}_{in}-\mathbf{k}_{out})\mathbf{r}_i} = \phi e^{-i\mathbf{k}\mathbf{r}_i}. \tag{3.45}$$

Now, the amplitude collected at $\mathbf{R}_d$ also includes a k dependent atomic form factor $f(k)$ which in our case we deliberately ignore.
The intensity collected at the detector is the square modulus of the total amplitude (structure factor of $S(\mathbf{k})$) where:

$$I \propto \left\langle \left| \sum_i A_i \right|^2 \right\rangle \propto |\phi|^2 \left\langle \sum_{ij} e^{i\mathbf{k}(\mathbf{r}_i-\mathbf{r}_j)} \right\rangle \propto N + \sum_{i\neq j} \left\langle e^{i\mathbf{k}(\mathbf{r}_i-\mathbf{r}_j)} \right\rangle = N + N(N-1) \left\langle e^{i\mathbf{k}(\mathbf{r}_i-\mathbf{r}_j)} \right\rangle = NS(\mathbf{k}).$$

The average may be computed as an integral over the generic pair correlation function, where the δ-like contribution for $k = 0$ may be removed by defining $h(r) = g(r) - 1$. And thus:

$$S(\mathbf{k}) = 1 + \frac{V}{N} \int \rho^{(2)}(\mathbf{0}, \mathbf{r}) e^{i\mathbf{k}\mathbf{r}} d\mathbf{r} = 1 + \rho \int g(r) e^{i\mathbf{k}\mathbf{r}} d\mathbf{r}. \tag{3.46}$$

Making a change of variable to polar coordinates yields:

$$\begin{aligned} S(k) = 1 + 2\pi\rho \int_{\text{range } r} \int_0^\pi r^2 g(r) \sin(\theta) e^{ikr\cos(\theta)} d\theta dr &= 1 + 2\pi\rho \int_{\text{range } r} r^2 g(r) \frac{e^{ikr} - e^{-ikr}}{ikr} dr \\ &= 1 + 4\pi\rho \int_{\text{range } r} g(r) \frac{r}{k} \sin(kr) dr \\ &\sim 1 + \rho V \delta(\mathbf{k}) + \frac{4\pi\rho}{k} \int_{\text{range } r} h(r) r \sin(kr) dr. \end{aligned}$$

3.4.3 Reversible Work Theorem

The core question in this subsection is to know how much work has to be made to bring two atoms in the fluid, initially far from each other, at a distance r.

Theorem 3.4.1 (Reversible Work Theorem)**.** *The required work is $W(r) = -k_B T \ln(g(r))$, where $g(r)$ is the pair correlation function defined in (3.44).*

Proof. The idea is to start from the usual definition for a reversible work and integrate it over a general path. In fact, it actually doesn't matter which path we consider and thus we will take a very simple path consisting of a straight line between two particles. The first particle will sit at the origin $\mathbf{0}$ and the second one will sit at the position $\mathbf{r}$, and thus a parametrization for our path will be of the form $r\hat{\mathbf{u}}$ where $r = |\mathbf{r}|$ and $\hat{\mathbf{u}}$ is an arbitrary direction vector (since the system is isotropic):

$$W(R) := -\int_{\text{path}} \langle \mathbf{f} \rangle \cdot d\mathbf{r} = -\int_{\infty}^{R} \langle \mathbf{f} \rangle \cdot \hat{\mathbf{u}} dr = -\int_{\infty}^{R} \hat{\mathbf{u}} \cdot \left\langle -\frac{\partial U(\mathbf{0}, \mathbf{r}, ...)}{\partial \mathbf{r}} \right\rangle_{|\mathbf{r}|=r} dr, \tag{3.47}$$

where we have used the fact that the force is $\mathbf{f} = -\frac{\partial U(\mathbf{0},\mathbf{r},...)}{\partial \mathbf{r}}$, for a potential $U(\mathbf{0}, \mathbf{r}, ...)$ with $\mathbf{0}$ and $\mathbf{r}$ representing the positions of the first and second particles (the remaining ones can be anywhere in the system). Now,

$$\begin{aligned}
\left\langle -\frac{\partial U(\mathbf{0}, \mathbf{r}, ...)}{\partial \mathbf{r}} \right\rangle_{|\mathbf{r}|=r} &= \int -\frac{\partial U(\mathbf{0}, \mathbf{r}, ...)}{\partial \mathbf{r}} \frac{e^{-\beta U(\mathbf{0},\mathbf{r},...)}}{Q} d\mathbf{r}_3 ... d\mathbf{r}_N \\
&= \frac{\int -\frac{\partial U(\mathbf{0},\mathbf{r},...)}{\partial \mathbf{r}} e^{-\beta U(\mathbf{0},\mathbf{r},...)} d\mathbf{r}_3 ... d\mathbf{r}_N}{\int e^{-\beta U(\mathbf{0},\mathbf{r},...)} d\mathbf{r}_3 ... d\mathbf{r}_N} \\
&= \frac{1}{\beta} \frac{\partial}{\partial \mathbf{r}} \ln \left(\int e^{-\beta U(\mathbf{0},\mathbf{r},...)} d\mathbf{r}_3 ... d\mathbf{r}_N \right) \\
&= \frac{1}{\beta} \frac{\partial}{\partial \mathbf{r}} \ln \left(\frac{N(N-1)}{\rho^2} \frac{1}{Q} \int e^{-\beta U(\mathbf{0},\mathbf{r},...)} d\mathbf{r}_3 ... d\mathbf{r}_N \right) \\
&= \frac{1}{\beta} \frac{\partial}{\partial \mathbf{r}} \ln \left(\frac{N(N-1)}{\rho^2} \mathbb{P}^{(2)}(\mathbf{r_1} = \mathbf{0}, \mathbf{r}_2 = \mathbf{r}) \right) = \frac{1}{\beta} \frac{\partial}{\partial \mathbf{r}} \ln \left(\frac{\rho^{(2)}(\mathbf{0}, \mathbf{r})}{\rho^2} \right),
\end{aligned}$$

where, in the fourth equality, we have used the fact that adding constants in the logarithm will not change anything upon differentiation. Remark that we have also used (3.43). Thus, since we have the following projection of the derivative $\hat{\mathbf{u}} \cdot \frac{\partial}{\partial \mathbf{r}} = \frac{\partial}{\partial r}$ and knowing that $\lim_{r\to\infty} g(r) = 1$, we finally obtain:

$$W(R) = -\frac{1}{\beta} \int_{\infty}^{R} \frac{\partial}{\partial r} \ln(g(r)) dr = -\frac{1}{\beta} \ln(g(R)) = -k_B T \ln(g(R)). \tag{3.48}$$

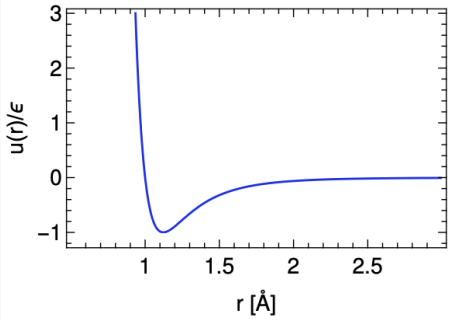
□

3.4.4 Pair Potentials and Thermodynamic Properties of Liquids

Consider the case in which $U(\{\mathbf{r}_i\}) = \frac{1}{2} \sum_{i \neq j} u(|\mathbf{r}_i - \mathbf{r}_j|)$ called a **pair potential**. We may compute explicitly the thermal average of the internal energy $\langle U \rangle$ based on the pair correlation function $g(r)$:

$$\begin{aligned}
\langle U \rangle = \int \sum_{i \neq j} u(|\mathbf{r}_i - \mathbf{r}_j|) \frac{e^{-\beta U}}{2Q} d\{\mathbf{r}_i\} = N(N-1) \int u(|\mathbf{r}_1 - \mathbf{r}_2|) \frac{e^{-\beta U}}{2Q} d\{\mathbf{r}_i\} &= \frac{1}{2} \int u(|\mathbf{r}|) \rho^{(2)}(\mathbf{0}, \mathbf{r}) d\mathbf{r} d\mathbf{r}' \\
&= \frac{1}{2} N \rho \int 4\pi r^2 g(r) u(r) dr.
\end{aligned}$$

Remark 3.4.2. *Remark that combined with the ideal gas term* $\langle K\rangle = \frac{3Nk_BT}{2}$ *gives the internal energy* $\langle E\rangle$.



Now write $W(r) = u(r) + \Delta W$, where we consider ΔW small in the **dilute limit** (i.e. as $\rho \to 0$). We can then write:

$$\frac{\langle U\rangle}{N} \sim \frac{\rho}{2}\int 4\pi r^2 u(r) e^{-\beta u(r)}(1+\mathcal{O}(\rho))dr = \frac{\langle E\rangle - \langle E_{\text{ideal gas}}\rangle}{N} = \frac{\Delta E}{N}. \tag{3.49}$$

Recall that the free energy A satisfies $\ln(Q) = -\beta A$ and $\langle E\rangle = -\frac{\partial}{\partial\beta}\ln(Q)$. Then:

$$\frac{\Delta E}{N} \equiv \frac{\Delta\langle E\rangle}{N} = \frac{\partial}{\partial\beta}\left(\beta\frac{\Delta A}{N}\right) = \frac{\rho}{2}\int 4\pi r^2 u(r)e^{-\beta u(r)}(1+\mathcal{O}(\rho))dr = \frac{\rho}{2}\int 4\pi r^2 u(r)e^{-\beta u(r)}dr + \mathcal{O}(\rho^2).$$

Integrating over β to obtain the free energy yields:

$$\begin{aligned}\beta\frac{\Delta A}{N} = \frac{\rho}{2}\int 4\pi r^2\int_0^\beta u(r)e^{-\alpha u(r)}d\alpha dr + \mathcal{O}(\rho^2) &= \frac{\rho}{2}\int 4\pi r^2\left[-e^{-\alpha u(r)}\right]_0^\beta dr + \mathcal{O}(\rho^2)\\ &= \frac{\rho}{2}\int 4\pi r^2\left(1-e^{-\beta u(r)}\right)dr + \mathcal{O}(\rho^2),\end{aligned}$$

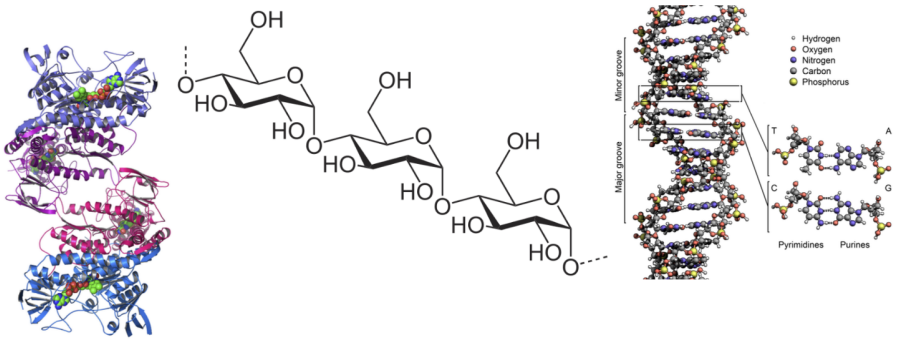
where we integrated from infinite temperature ($\beta = 0$) such that the system behaves as an ideal gas. Now, recall that $\rho = \frac{N}{V}$ such that $p := -\frac{\partial A}{\partial V} = -N\frac{\partial(A/N)}{\partial\rho}\frac{\partial\rho}{\partial V} = -N\left(-\frac{N}{V^2}\right)\frac{\partial(A/N)}{\partial\rho} = \rho^2\frac{\partial(A/N)}{\partial\rho}$. Hence, considering $A = A_{\text{ideal gas}} + \Delta A$ where ΔA is small in the dilute limit, one finds that the equation of state βp is:

$$\beta p = -\beta\frac{\partial A}{\partial V} = -\beta\frac{\partial A_{\text{ideal gas}}}{\partial V} - \beta\frac{\partial\Delta A}{\partial V} = (\beta p)_{\text{ideal gas}} + \beta\rho^2\frac{\partial\left(\frac{\Delta A}{N}\right)}{\partial\rho} = \rho + \rho^2\underbrace{\int 2\pi r^2\left(1-e^{-\beta u(r)}\right)dr}_{\text{second virial coefficient}} + \mathcal{O}(\rho^3),$$

where we have used the fact that the equation of state of an ideal gas is $\beta p = \rho$ along with the expression for $\beta\frac{\Delta A}{N}$. The coefficient in ρ^2 is called the *second virial coefficient*, and it corresponds to the leading order correction for the equation of state βp of an ideal gas.

3.5 Statistical Mechanics of Polymers

Polymers are ubiquitous in Biology as well as in Materials Science. They are repeated units of monomers, typically joined by stiff covalent bond but with relatively flexible chain conformation.



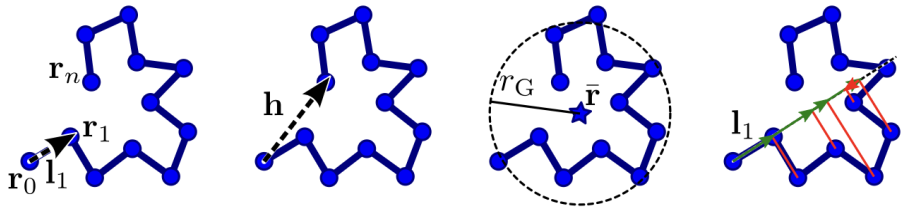
3.5.1 Characterizing Polymer Configurations and some General Relations

To characterize polymer configurations we use a *coarse-grained* representation, where each monomer is a single site (i.e. a bead) with position $\mathbf{r}_i$ for $i \in \{0,...,n\}$. Vectors joining pairs of nodes are labeled as $\mathbf{l_i} = \mathbf{r}_i - \mathbf{r}_{i-1}$ where $l = \sqrt{\langle \mathbf{l} \cdot \mathbf{l} \rangle}$. Knowledge of the position of each node matter relatively little hence we can use average properties such as the end-to-end vector $\mathbf{h} = \mathbf{r}_n - \mathbf{r}_0$.
A more global picture of the size of the polymer is given by its **gyration radius**:

$$r_G^2 = \frac{1}{n+1}\sum_{i=0}^{n}(\mathbf{r}_i - \overline{\mathbf{r}})^2 \in \mathbb{R}, \tag{3.50}$$

where $\overline{\mathbf{r}} = \frac{1}{n+1}\sum_i \mathbf{r}_i$. Furthermore, a characterization of the polymer's rigidity is given by the **persistence length**:

$$\xi = \frac{1}{l}\sum_{j=0}^{\infty}\mathbf{l}_i \cdot \mathbf{l}_{i+j}. \tag{3.51}$$



Now, the end-to-end vector is generally not very interesting since $\langle \mathbf{h} \rangle = \sum_i \langle \mathbf{l}_i \rangle = 0$ due to the symmetry. However, a more useful quantity is given by the **end-to-end distance** since it contains information about the extent of the polymer. Recall that $\mathbf{r}_n = \mathbf{r}_0 + \sum_{i=1}^{n}\mathbf{l}_i$ and then:

$$h^2 = \langle |\mathbf{r}_n - \mathbf{r}_0|^2 \rangle = \left\langle \left| \sum_{i=1}^{n} \mathbf{l}_i \right|^2 \right\rangle = \left\langle \sum_{i,j=1}^{n} \mathbf{l}_i \cdot \mathbf{l}_j \right\rangle = \sum_{i,j=1}^{n} \langle \mathbf{l}_i \cdot \mathbf{l}_j \rangle = nl^2 + 2\sum_{i=1}^{n}\sum_{j>i}\langle \mathbf{l}_i \cdot \mathbf{l}_j \rangle. \tag{3.52}$$

Now, the gyration radius is closer to experimental observables but harder to compute. In fact it can be shown that r_G^2 is related to h^2. Since life is short we will not delve into the proof of the following approximation, however assuming that the persistence length consists of a single bond such that $\xi = l$ one should find,

$$r_G^2 = \frac{1}{n+1}\sum_{i=0}^{n}(\mathbf{r}_i - \overline{\mathbf{r}})^2 = \frac{(n+2)}{6(n+1)}nl^2 + \sum_{i=1}^{n}\sum_{j>i}\frac{i(n+1-j)}{(n+1)^2}\langle \mathbf{l_i} \cdot \mathbf{l_j} \rangle \approx \frac{nl^2}{6} \sim \frac{h^2}{6}. \tag{3.53}$$

We also remark that the persistence length ξ can be related to h. Indeed, starting from an arbitrary node i we have:

$$\xi = \frac{1}{l}\sum_{j=i}^{\infty}\langle \mathbf{l}_i \cdot \mathbf{l}_j\rangle \approx \frac{1}{2nl}\sum_{i=1}^{n}\left(\sum_{j=i}^{n}\langle \mathbf{l}_i \cdot \mathbf{l}_j\rangle + \sum_{j=1}^{i}\langle \mathbf{l}_i \cdot \mathbf{l}_j\rangle\right) = \frac{1}{2nl}\left(\sum_{i=1}^{n}\langle \mathbf{l}_i^2\rangle + \sum_{i=1}^{n}\langle \mathbf{l}_i \cdot \mathbf{l}_j\rangle\right) = \frac{h^2}{2nl} + \frac{l}{2}, \tag{3.54}$$

where we have multiplied the sum over i by $\frac{1}{2}$ to avoid double counting the sums over j (they are equal by symmetry), and we have used the facts that $\sum_{i=1}^{n}\langle \mathbf{l}_i^2\rangle = nl^2$ and $\sum_{i=1}^{n}\langle \mathbf{l}_i \cdot \mathbf{l}_j\rangle = h^2$.

3.5.2 Freely Jointed Chain

Let us now look at the simplest model for the characterization of a polymer by considering a chain as a random walk in 3D with steps of equal length l. In this model, the orientation of each segment is independent from that of the others, thus for the correlation $\langle \mathbf{l}_i \cdot \mathbf{l}_j\rangle = l^2\delta_{ij}$. As a result, the persistent length is just one bond,

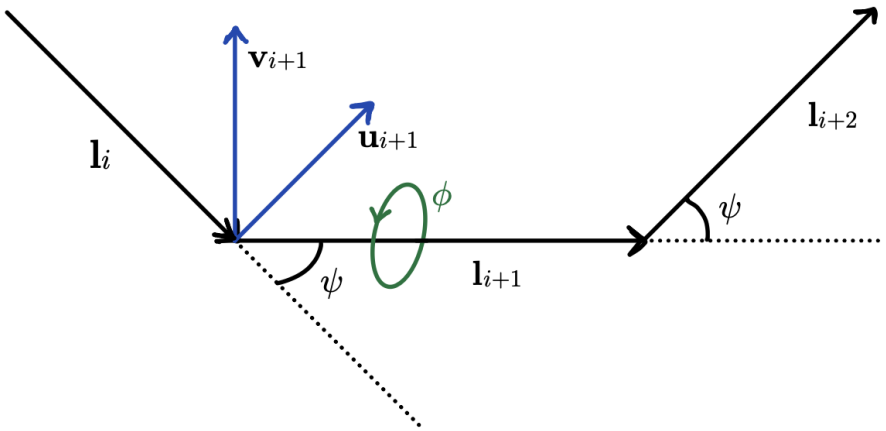
$$\xi = \frac{1}{l}\sum_{j=i}^{\infty}\langle \mathbf{l}_i \cdot \mathbf{l}_j\rangle = \frac{1}{l}\sum_{j=1}^{\infty} l^2\delta_{ij} = l. \tag{3.55}$$

Also, we have that the end-to-end distance (3.52) obeys a simple scaling law, $h^2 = nl^2 \Longrightarrow h = l\sqrt{n}$. Finally, for the gyration radius we again omit the proof of this result, nevertheless we find:

$$r_G^2 = \frac{1}{(n+1)^2}\sum_i\sum_{j>i}(j-i)l^2 = \frac{(n+2)}{6(n+1)}nl^2 \xrightarrow{n\to\infty} \frac{h^2}{6}. \tag{3.56}$$

3.5.3 Freely Rotating Chain

We now study a slightly different model where we use the general fact that often the joint angles are rather rigid. Thus, these angles ψ between different bonds may be taken to be constant so that $\langle \mathbf{l}_i \cdot \mathbf{l}_{i+1}\rangle = l^2\cos(\psi)$. Let us now see how we can compute $\langle \mathbf{l}_i \cdot \mathbf{l}_{i+2}\rangle$.



As we can see from the above representation, general correlations can be found iteratively by considering a sum over orthogonal unit vectors, one of which is parallel to intermediate bonds. Here, the orthogonal unit vectors are $\mathbf{u}_{i+1}$ and $\mathbf{v}_{i+1}$. Since integrals over orthogonal projections on bonds average to zero by symmetry, for $\langle \mathbf{l}_i \cdot \mathbf{l}_{i+2}\rangle$ we obtain:

$$\langle \mathbf{l}_i \cdot \mathbf{l}_{i+2}\rangle = \langle \mathbf{l}_i \cdot \widehat{\mathbf{l}}_{i+1}\rangle\langle \mathbf{l}_{i+2} \cdot \widehat{\mathbf{l}}_{i+1}\rangle + \langle \mathbf{l}_i \cdot \mathbf{u}_{i+1}\rangle\langle \mathbf{l}_{i+2} \cdot \mathbf{l}_{i+1}\rangle + \langle \mathbf{l}_i \cdot \mathbf{v}_{i+1}\rangle\langle \mathbf{l}_{i+2} \cdot \mathbf{v}_{i+1}\rangle, \tag{3.57}$$

where we denoted by $\widehat{\mathbf{l}}_{i+1}$ the unit vector in the direction of $\mathbf{l}_{i+1}$ being orthogonal to $\mathbf{u}_{i+1}$ and $\mathbf{v}_{i+1}$. Now, the last two terms yield zero for the reason that we have mentioned previously, and then we finally obtain:

$$\begin{aligned}\langle \mathbf{l}_i \cdot \mathbf{l}_{i+2} \rangle &= \langle \mathbf{l}_i \cdot \widehat{\mathbf{l}}_{i+1} \rangle \langle \mathbf{l}_{i+2} \cdot \widehat{\mathbf{l}}_{i+1} \rangle + \cancelto{0}{\langle \mathbf{l}_i \cdot \mathbf{u}_{i+1} \rangle \langle \mathbf{l}_{i+2} \cdot \mathbf{l}_{i+1} \rangle} + \cancelto{0}{\langle \mathbf{l}_i \cdot \mathbf{v}_{i+1} \rangle \langle \mathbf{l}_{i+2} \cdot \mathbf{v}_{i+1} \rangle} \\ &= \Big(|\mathbf{l}_i| |\widehat{\mathbf{l}}_{i+1}| \cos(\psi) \Big) \Big(|\mathbf{l}_{i+2}| |\widehat{\mathbf{l}}_{i+1}| \cos(\psi) \Big) \\ &= l^2 \cos^2(\psi).\end{aligned}$$

As a general rule, it can be shown that for any k, $\langle \mathbf{l}_i \cdot \mathbf{l}_{i+k} \rangle = l^2 \cos^k(\psi)$. Note that if $\psi = \frac{\pi}{2}$ then this chain reduces to the case of the freely jointed chain presented in the previous subsection (same persistence length). Now, for the persistence length ξ in this case we find:

$$\xi = \frac{1}{l} \sum_{j=i}^{\infty} \langle \mathbf{l}_i \cdot \mathbf{l}_j \rangle \overset{k=j-i}{=} \frac{1}{l} \sum_{k=0}^{\infty} \langle \mathbf{l}_i \cdot \mathbf{l}_{i+k} \rangle = l \sum_{k=0}^{\infty} \cos^k(\psi) = \frac{l}{1 - \cos(\psi)}. \tag{3.58}$$

For the end-to-end distance h^2 we can use (3.54) to find:

$$\xi \approx \frac{h^2}{2nl} + \frac{l}{2} \Longrightarrow h^2 \approx 2nl\xi - nl^2 = \frac{2nl^2}{1 - \cos(\psi)} - nl^2 = nl^2 \left(\frac{1 + \cos(\psi)}{1 - \cos(\psi)} \right). \tag{3.59}$$

Again note that for $\psi = \frac{\pi}{2}$ we recover the end-to-end distance of the freely jointed polymer. Finally, we can look at a quantity called the *polymer stiffness* which is defined as:

$$C_\infty := \frac{h^2}{h^2_{\text{Freely Jointed Chain}}} = \frac{1 + \cos(\psi)}{1 - \cos(\psi)}, \tag{3.60}$$

and it indicates how a chain is behaving differently from a freely jointed polymer/chain.

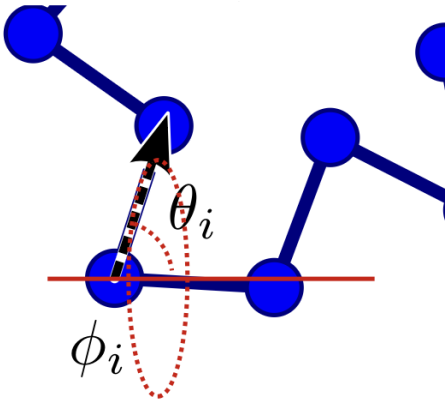
3.5.4 Partition Function at Constant f and Polymer Extension

Consider an experiment/model in which a constant force f is applied to the end-points of a polymer (e.g. optical tweezers, AFM,...). For this constant force f, the work done to extend the polymer to an extension x is fx (equivalent to the pV term!). Thus, the equivalent of the NpT ensemble here is nfT. We will assume that since we are considering a freely jointed chain then there is no difference in the internal energy of different conformations of the polymer (different shapes of a polymer due to the rotational freedom around single bonds) and thus we may set the energy $E_\nu = cst. = 0$. The partition function then reads,

$$\Delta(n, f, T) = \sum_\nu e^{-\beta E_\nu + \beta f x_\nu} = \sum_\nu e^{\beta f x_\nu}, \tag{3.61}$$

where $x_\nu = \mathbf{x}_\nu \cdot \widehat{\mathbf{x}}$ is the projection of the end-to-end distance for a certain configuration ν of the polymer along the x-axis (i.e. along the direction $\widehat{\mathbf{x}}$). In simpler words, x_ν represents the total extension of the polymer along the x-axis (i.e. the full length of the chain).

Now, since in a freely jointed chain the orientation of the bonds in 3D space is arbitrary, we can choose to describe the bonds and thus the microstates of the polymer in spherical coordinates with polar angles $\{\theta_i\}$ and azimuthal angles (angles of rotation) $\{\phi_i\}$. The orientation with respect to the x-axis will be represented by θ_i and the one of the plane orthogonal to the x-axis (red line below) by ϕ_i. Note that $\theta_i \neq \psi_i$!



Then, the microstates of the polymer can be described with:

$$\sum_{\nu} \square = \int_0^{2\pi} d\phi_1 ... \int_0^{2\pi} d\phi_n \int_0^{\pi} d\theta_1 \sin(\theta_1) ... \int_0^{\pi} d\theta_n \sin(\theta_n) \square. \tag{3.62}$$

Also, the extension of the polymer along x can be computed as $x_\nu \equiv \sum_i \mathbf{l}_i \cdot \hat{\mathbf{x}} \equiv \sum_{i=1}^n l\cos(\theta_i)$. Since all the polymer bonds are independent, the expression for Δ factors out,

$$\begin{aligned}
\Delta(n,f,T) = \prod_{i=1}^{n} \int_0^{\pi} \int_0^{2\pi} e^{\beta f x_\nu} \sin(\theta_i) d\phi_i d\theta_i &= \prod_{i=1}^{n} 2\pi \int_0^{\pi} e^{\beta f l \cos(\theta_i)} \sin(\theta_i) d\theta_i \\
&= \prod_{i=1}^{n} \left[-\frac{2\pi}{\beta f l} \left(e^{\beta f l \cos(\theta_i)} \right) \right]_0^{\pi} \\
&= \prod_{i=1}^{n} \left[\frac{4\pi}{\beta f l} \left(e^{\beta f l} - e^{-\beta f l} \right) \right] = \left[\frac{4\pi}{\beta f l} \sinh(\beta f l) \right]^n.
\end{aligned}$$

From there, we can derive the mean extension of the polymer along x from Δ,

$$\begin{aligned}
\langle x \rangle = \frac{\partial \ln(\Delta)}{\partial(\beta f)} = \frac{\partial}{\partial(\beta f)} \left[n \ln\left(\frac{4\pi}{\beta f l} \sinh(\beta f l) \right) \right] &= n \frac{\left(-\frac{4\pi}{(\beta f)^2 l} \sinh(\beta f l) + \frac{4\pi}{\beta f l} l \cosh(\beta f l) \right)}{\left(\frac{4\pi}{\beta f l} \sinh(\beta f l) \right)} \\
&= nl \left(\coth(\beta f l) - \frac{1}{\beta f l} \right).
\end{aligned}$$

- Now, using $\coth(x) = \frac{1}{x} + \frac{x}{3} + \mathcal{O}(x^3)$ and for small f we find the approximation,

$$\langle x \rangle = nl \left(\coth(\beta f l) - \frac{1}{\beta f l} \right) = nl \left(\frac{\beta f l}{3} + \mathcal{O}[(\beta f l)^3] \right) \approx \frac{nl^2\beta}{3} f \Longrightarrow f \approx \frac{3}{nl^2\beta} \langle x \rangle. \tag{3.63}$$

Thus, for small f, $f \propto \langle x \rangle$ which suggests that we have the behavior of an oscillator. However, recall that we assumed that the internal energy of each configuration E_ν was equal to zero. Hence this is an oscillator that is entirely driven by entropy!

- For large f now we have that $\coth(\beta f l) \to 1$, then,

$$\langle x \rangle \approx nl \left(1 - \frac{1}{\beta f l} \right) \Longrightarrow f \approx \frac{n}{\beta} \frac{1}{nl - \langle x \rangle}. \tag{3.64}$$

Thus, since the polymer once fully extended has a size of nl then by the above relation we conclude that the force diverges when the polymer approaches its full extension, i.e. when the extension approaches $\langle x \rangle = nl$.